

Decomposition and framing of F-bundles, machine-checked: a Lean 4 formalization published with its proofs withheld

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Abstract

We report on a Lean 4 / Mathlib formalization of the linear-algebraic core of *Decomposition and framing of F-bundles and applications to quantum cohomology* (arXiv:2411.02266), together with the machinery used to publish it. The library contains 692 kernel-checked declarations (539 theorems and 153 definitions) in 30 modules, builds without `sorry`, and every declaration’s axiom closure is contained in `Classical.choice`, `Quot.sound`, `propext`. Formalizing the paper turned up one statement that is false as written — the vanishing-diagonal lemma behind the projective-bundle application — for which we give a machine-checked counterexample and a corrected statement that suffices for the application.

This document is deliberately asymmetric. It *describes* the proofs but does not contain them: not one proof term, and no fragment of one, is printed here. What it does contain is the cryptographic object that stands in for them. Every declaration is re-submitted to the Lean kernel at extraction time and serialized; the serialization is hashed; the digest is committed to with a Pedersen commitment; and a Fiat–Shamir compiled Schnorr proof of knowledge of the opening, bound to the statement it belongs to, is published next to the statement in clear. All 692 statements, all 692 commitments and all 692 transcripts are reproduced in this document and in its companion volume, and can be checked with nothing but a SHA-256 implementation and modular exponentiation. The 159.3 MB of proof terms they refer to are never published. The completeness, hiding, soundness and binding properties of exactly this publication scheme are themselves theorems of the formalized library.

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1 What this document is

There are two things one may want from a formalization: to know *what* has been machine-checked, and to be able to re-check it. The first needs only the statements. The second needs the proofs — or a reason to believe someone holds them.

This document takes the first seriously and replaces the second by a cryptographic commitment. Its three layers are:

- (i) **a description of the mathematics that was formalized** (§3), written so that a reader of arXiv:2411.02266 can see which of its results, and in what generality, are now machine-checked;
- (ii) **the statements themselves, in clear** — the Lean types, pretty-printed unambiguously, for the principal results in §6.3, for every declaration in the companion volume `ledger-appendix.pdf`;
- (iii) **a proof of possession of the proofs** (§5, §6): per-declaration Pedersen commitments to the digests of the withheld kernel data, each with a zero-knowledge proof of knowledge of the opening bound to its statement, and an aggregate over the whole library.

What is not here. No proof term, no tactic script, no proof sketch of any formalized lemma. Where the argument behind a statement matters for reading it, we say in words what shape the argument takes — an induction on the total degree, a Neumann series, a Hensel lift — but the artefacts that Lean’s kernel accepted stay with the issuer. Section 5.7 states precisely in what sense they stay hidden, including a machine-checked perfect-hiding theorem and an empirical leak test over everything published.

2 The source paper and the scope of the formalization

The source is T. Hinault, T. Y. Yu, C. Zhang and S. Zhang, *Decomposition and framing of F-bundles and applications to quantum cohomology*, arXiv:2411.02266. An F-bundle over a base B is, in the trivialized picture used throughout, a bundle over $B \times D_u$ with a connection having a pole along $u = 0$,

$$\nabla_{u\partial_u} = u\partial_u + u^{-1}K + G, \quad \nabla_{\partial_t} = \partial_t + u^{-1}T,$$

subject to regularity conditions in u and to flatness. The paper's two main theorems are a *spectral decomposition* — a bundle whose K has, in a suitable sense, disjoint spectra splits, compatibly with the connection — and an *extension of framing* — a framing given over a subspace of the base extends, uniquely, over the whole base. The applications compute quantum cohomology of projective bundles and blowups.

The global statements are phrased over formal schemes and non-archimedean analytic spaces. Mathlib does not have that geometry, so those statements are out of scope here. What is in scope, and is what this library contains, is the linear algebra that carries the proofs: the classification of framed F-bundles over a point, the Sylvester theory behind the splitting, the spectral decomposition of a connection on the formal disk, the two matrix lemmas that drive the extension of framing (in the multi-variable generality in which the paper uses them), the framing recursions in the ordinary and logarithmic directions, the gauge covariance that makes the induction step legitimate, and the matrix input of the projective-bundle application. Table 1 lists the modules.

3 The mathematics that is machine-checked

This section describes the content. It is a guide to the statements, which are published in clear; the proofs are withheld, and every claim below is backed by the corresponding entry of the statement ledger.

3.1 Framed F-bundles over a point

For a trivialized rank- m bundle over $\mathbb{k}[[u]]$ carrying two framed connections $u\partial_u + u^{-1}K + G$ and $u\partial_u + u^{-1}K' + G'$, a gauge equivalence is an invertible $\Phi(u)$ over $\mathbb{k}[[u]]$ with

$$u(u\partial_u\Phi) + K\Phi + uG\Phi = \Phi K' + u\Phi G'.$$

Expanding $\Phi = \sum_k P_k u^k$ turns this into the coefficientwise system

$$KP_0 = P_0K', \quad kP_k + KP_{k+1} + GP_k = P_{k+1}K' + P_kG' \quad (k \geq 0).$$

With $K = \text{diagonal } \xi$ and ξ injective — the simple-eigenvalue case, written in an eigenbasis — the library proves that for every invertible Q with $KQ = QK'$ and $G_{ii} = (QG'Q^{-1})_{ii}$ there is exactly one gauge transformation with $\Phi|_{u=0} = Q$; that those two conditions are also necessary; and hence the classification, that the two framed connections are gauge equivalent if and only if such a Q exists. The same statements are given for matrices over $\mathbb{k}[[u]]$, where invertibility in $\text{GL}(m, \mathbb{k}[[u]])$ is characterized by the constant term.

The engine is the normalized recursion $[K, R_{k+1}] = R_k G'' - GR_k - kR_k$ with $R_0 = 1$: the operator $[\text{diagonal } \xi, -]$ has the diagonal matrices as kernel and the matrices with vanishing diagonal as image, so each order is solvable, and the diagonal ambiguity left at each step is fixed uniquely — which is where characteristic zero enters, exactly as in the paper.

3.2 Sylvester operators, disjoint spectra and block splitting

The paper's splitting hypothesis is that distinct blocks of K have disjoint spectra. The library renders it as coprimality of characteristic polynomials, and proves that over an algebraically closed field the two conditions agree. Over an arbitrary commutative ring, coprimality of χ_A and χ_B makes $X \mapsto AX - XB$ injective (Cayley–Hamilton makes $\chi_B(A)$ invertible); over a field it makes it bijective, so $AX - XB = C$ is uniquely solvable.

Two consequences are the paper's block lemmas, and both are formalized for a decomposition of the index set into arbitrarily many blocks, not just two: if U is block diagonal with distinct blocks of coprime characteristic polynomials and $[U, X]$ is block diagonal, then X is block diagonal; and every purely off-diagonal C is $[U, X]$ for exactly one purely off-diagonal X . From these follow the splitting of the base directions along the decomposition, and one step of the splitting in the u -direction.

3.3 Spectral decomposition on the formal disk

Assembling the block lemmas order by order gives the spectral decomposition itself, in a matrix form and in a power series form. For K block diagonal with distinct blocks of coprime characteristic polynomials, every connection $u\partial_u + u^{-1}K + A(u)$ is gauge equivalent, by a Φ with $\Phi|_{u=0} = 1$ and all higher coefficients purely off-diagonal, to one whose connection form is block diagonal at every order — and that Φ , with the resulting connection form, is unique. At each order the block diagonal part of the obstruction is absorbed into the new connection form and its off-diagonal part is removed by the Sylvester step; without the normalization, uniqueness fails precisely by a block diagonal gauge transformation.

3.4 The two matrix lemmas

The extension of framing rests on two statements about families $(X_k)_k$ of matrices over a formal power series ring. In the ordinary directions, $\partial_{t_i} X_k = -[T_i, X_{k+1}]$; in the logarithmic directions, $q_i \partial_{q_i} X_k = [Q_i, X_{k+1}]$. The library proves, over $\mathbb{k}[[t_i, i \in \sigma]]$ with σ an arbitrary index type, that such a family is determined by its values at the origin, and vanishes when they vanish — the logarithmic case under the paper’s nilpotency condition on the adjoint action of Q_i at the origin. Both proofs are strong inductions on the total degree of a monomial; in the logarithmic case the induction step iterates the adjoint action until nilpotency kills it.

Three further modules pin the statement down. The one-variable formulation is proved to be literally the $\sigma = \text{Unit}$ instance of the multivariate one, with no drift in the hypotheses. Both lemmas are reproved over an arbitrary commutative $\mathbb{Q}$ -algebra, which is the generality the paper needs when the remaining directions of the base are hidden in the coefficient ring. And two counterexamples record that neither hypothesis can be dropped: with $T = 0$ a nonzero constant family solves the ordinary equations, and for $Q = \text{diag}(0, 1)$, whose adjoint action is not nilpotent, the nonzero family $X_k = (-1)^k q E_{12}$ solves the logarithmic ones and vanishes at the origin.

3.5 Existence, and why several directions are different

Uniqueness has a companion: in one formal direction, for *any* T and *any* prescribed initial values, a solution of $\partial_t X_k = -[T, X_{k+1}]$ exists, so the formal Cauchy problem is well posed. With several directions the system is overdetermined, and the library makes the obstruction explicit: any solution satisfies $[[T_i, T_j], X_k(0)] = 0$, and a concrete two-variable instance with $T_1 = E_{11}$, $T_2 = E_{12}$ admits no solution with the prescribed values at the origin, while the same prescription is realized in one variable.

3.6 Framing recursions and the nilpotency condition

Framing the connection $u^{-1}T$ in the direction ∂_t means solving $u \partial_t P + TP = P\tilde{T}$ with $P|_{t=0} = 1$ and $\tilde{T}$ independent of u ; in a logarithmic direction it means $u q \partial_q P + QP = P\tilde{Q}$ with $P|_{q=0} = 1$. The library builds both solutions by the paper’s recursions and proves existence *and* uniqueness, in the coefficientwise form and in the form of an equation over $R[[t]][[u]]$, resp. $R[[q]][[u]]$, over an arbitrary commutative $\mathbb{Q}$ -algebra R .

The two cases differ in an essential way. In the ordinary direction the definition of $\tilde{T}_\ell$ is exactly what makes the u -constant part of the order- ℓ equation vanish, so that the rest is divisible by u and defines $P_{\ell+1}$. In the logarithmic direction the unknown P_ℓ occurs on both sides at the same order through the commutator with Q_0 ; the operator to invert is $\ell + \text{ad}(Q_0) \circ \text{shift}$, and it is inverted by a *finite* Neumann series — which terminates only because $\text{ad}(Q_0)$ is nilpotent. That nilpotency is not assumed but derived: from the flatness relations of a strong framing one gets $[G, Q] = Q$, hence $[G, Q^n] = n Q^n$, so in characteristic zero a non-nilpotent Q would give $\text{ad}(G)$ infinitely many eigenvalues.

3.7 Gauge covariance and one step of the extension of framing

The induction that extends a framing frames one direction at a time by a gauge transformation and then feeds the flatness equations into the matrix lemmas; for that the curvature must be gauge covariant. The library proves this in the generality in which it is used: an arbitrary, possibly noncommutative ring with two commuting derivations given as additive maps with explicit Leibniz hypotheses, where

$$\text{gauge}(X) = q(dp + Xp), \quad \text{curv}(X_1, X_2) = d_1X_2 - d_2X_1 + [X_1, X_2],$$

and curv of the gauged pair is $q \cdot \text{curv}(X_1, X_2) \cdot p$; in particular flatness is preserved. It then checks that the operators $u\partial_t$ and $uq\partial_q$ occurring in the framing equations really are derivations in that sense, identifies the flatness predicates with the vanishing of the curvature, and combines the two halves into one step of the extension of framing: given a flat pair of directions, a gauge transformation framing the first, and the framing of the second at $t = 0$ (resp. $q = 0$, under the nilpotency condition), the second direction is framed as well. Vacuity checks accompany the statement: the hypotheses are satisfiable with a nonzero connection matrix, and the hypothesis at the origin cannot be dropped.

3.8 The projective-bundle application, and a lemma that is false as stated

For the quantum cohomology of a projective bundle $\mathbb{P}(V) \rightarrow X$ with V of rank $m+1$, the relevant operator acts by a companion matrix: the cyclic shift corrected in its last column by the Chern classes of V . The paper's argument needs two inputs, and both are formalized: the companion matrix is diagonalizable, its eigenvalues being the $(m+1)$ -st roots of unity deformed by the Chern classes (proved by Hensel lifting along the nilpotent ideal generated by the Chern classes, with the eigenvector matrix inverted by a Lagrange identity); and the diagonal entries of a certain conjugate vanish.

The second input, the vanishing-diagonal lemma, is *false as stated* for a nontrivial bundle: its proof uses cyclic relations that hold only modulo the Chern classes. The library records a machine-checked counterexample — $m = 2$ over the dual numbers $\mathbb{Q}[\varepsilon]$ with $c_1 = \varepsilon$, $c_2 = 0$, which is the shape of the cohomology of $X = \mathbb{P}^1$ with $V = \mathcal{O} \oplus \mathcal{O}(1)$; for $\mu = (1, -1)$ the first diagonal entry is $\varepsilon/2 \neq 0$, and this is independent of the diagonalizing matrix chosen — together with the corrected statement, that the diagonal entries lie in every ideal containing the Chern classes, i.e. vanish modulo $(c_1, \dots, c_m)$. The corrected form is what the application uses. For the trivial bundle the lemma holds as stated, and that case is formalized too.

3.9 Conventions and deviations

- The base field is any field of characteristic zero; the framing modules are stated over an arbitrary commutative $\mathbb{Q}$ -algebra, the generality in which the paper applies them when framing one direction after another.
- In the classification over a point, the conjugating matrix Q is the value of the gauge transformation at $u = 0$, the inverse of the matrix φ of the paper's phrasing; with that convention the two stated conditions are consistent with the recursion.
- Block decompositions are formalized for an arbitrary partition of the index set; two earlier modules treat the two-block case separately.
- The geometric statements — decomposition and extension of framing as statements about F-bundles over a formal or non-archimedean base, and the quantum-cohomology applications — remain out of scope, as does everything that would need that geometry.

4 The library as an artifact

- 692 declarations are extracted, 539 of them theorems, in 30 modules; the whole library builds with no `sorry` and no `admit`, and no `axiom` declaration is introduced.
- Every declaration’s axiom closure is contained in `Classical.choice`, `Quot.sound`, `propext` — Lean’s standard axioms, and nothing else. In particular no declaration depends on `sorryAx`.
- All 692 of the 692 declarations were re-submitted to the Lean kernel at extraction time, under a fresh name, and accepted: a second, independent kernel type-check of each term.
- The withheld kernel proof terms total 159 288 447 bytes (159.3 MB); the largest single term is 5 712 070 bytes and the median 12 863. The published statements total 2 741 330 bytes.
- A repository harness re-runs the build, the source gates, the axiom closure, the vacuity examples, an executed numeric benchmark of the gauge recursion, and a check that every module is covered; continuous integration runs it on every push.

Beside the library the repository carries explanatory material generated from the library’s own data — a dependency-graph and recursion simulator, and a set of figures — so that the quantitative claims made about the formalization are read from the compiled environment rather than typed by hand. The same discipline governs this document: every number above is read from the published artifacts by the generator that produced the tables.

5 Publishing a proof without publishing it

5.1 The problem

A formalization is convincing because a machine checked it. Publishing the statements alone tells the reader what was checked but gives no evidence that anything was; publishing the proofs gives the evidence but gives away the artifact. The pipeline described here takes the middle route: statements in clear, proofs withheld, and a cryptographic proof that the issuer holds proofs the kernel accepted for exactly those statements.

5.2 Extraction

A Lean program runs inside the compiled environment. For every declaration it reads the kernel type and, when there is one, the kernel proof term; re-submits both to the kernel under a fresh name, recording the verdict; computes the axiom closure; and writes the raw serialization to a witness file. The digest of the declaration is the SHA-256 of that serialization. The witness files are the secret this pipeline exists not to publish. The file name of a declaration is an injective escaping of its Lean name, so two declarations can never be committed to through the same witness, and the builders refuse to run if that ever fails.

5.3 Commitments and proofs of knowledge

Work in the subgroup of quadratic residues of the 2048-bit MODP group of RFC 3526, of prime order $q = (p - 1)/2$, with $g = 2^2$ and h derived by a nothing-up-my-sleeve hash procedure, so that no discrete logarithm relating g and h is known. For a declaration D with digest d_D , set $m_D = d_D \bmod q$, draw r_D uniformly, and publish the Pedersen commitment

$$C_D = g^{m_D} h^{r_D} \bmod p$$

together with a Fiat–Shamir compiled Schnorr proof of knowledge of the opening (m_D, r_D) : the prover draws a, b , sends $t = g^a h^b$, computes the challenge $c = H(\text{group}, C_D, t, \text{context}_D)$ and

answers $z = a + cm_D$, $w = b + cr_D \pmod{q}$. The verifier accepts when C_D and t lie in the prime-order subgroup and

$$g^z h^w = t \cdot C_D^c \pmod{p}.$$

The context binds the transcript to the declaration and to the digest of its published statement, so a transcript cannot be moved to another statement. The aggregate $C = \prod_D C_D$ is a commitment to the sum of the digests and carries its own proof.

5.4 The statement ledger

A commitment to a digest proves possession of *something*; the reader also has to know what. A statement is not a proof — the type of a declaration can be published in full without revealing a step of the term inhabiting it — and that asymmetry is what the ledger exploits. For every declaration the ledger carries: the statement in clear, pretty-printed with full names and universes; the SHA-256 of that published text and the SHA-256 of the kernel serialization of the type; the Pedersen commitment to the digest of the *hidden* proof term and nothing else about it; the Schnorr transcript, with its challenge bound to the statement; the axiom closure, the kernel replay verdict, and the size in bytes of the withheld term. Leaves pairing each statement with its commitment are hashed into a Merkle tree whose root is signed.

5.5 Distribution

Two Merkle trees (over the declaration records, and over the images of each bundle) with domain-separated leaf and node hashing, Ed25519 signatures on every certificate, manifest and per-image provenance record, and a payload — certificate, zero-knowledge proofs, ledger, keys, manifest, instructions, protocol and verifier — split across the images of a deck and hidden one bit per colour channel, so that collecting the images of a deck reconstructs a runnable copy of the proof object. The issuer’s signing key is additively split into 24 shares, one per bundle, each encrypted under a Paillier public key whose private key is itself XOR-split one piece per bundle: all 24 bundles are necessary and sufficient to recover it.

5.6 The scheme’s own correctness, machine-checked

The cryptographic layer is not merely tested. Its mathematical content is formalized in the same library, and those theorems are in the ledger like the rest:

theorem	what it says
<code>Sigma.sigma_complete</code>	an honest prover always convinces the verifier
<code>Sigma.sigma_special_soundness</code>	two accepting transcripts with different challenges extract the same randomness
<code>Sigma.sigma_hvzk_bijective</code>	simulated and real transcripts are in bijection
<code>Sigma.pedersen_binding, pedersen_hiding</code>	binding and hiding of the commitment
<code>Ledger.honestView_valid</code>	the entry an honest issuer publishes is accepted
<code>Ledger.honestView_eq_of_commit_eq</code>	two issuers holding proof terms with <i>different</i> digests publish the same entry
<code>Ledger.exists_unique_view_match</code>	for every alternative digest exactly one randomness produces the same entry: the entry is perfectly hiding for the proof term
<code>Ledger.extract_opening_of_two_views</code>	two accepted entries differing only in the challenge yield the same opening: the issuer knows the digest
<code>Ledger.collision_of_ledger_records</code>	a statement cannot be swapped next to its commitment without an explicit hash collision
<code>Merkle.foldPath_eq_root, collision_of_foldPath_eq</code>	completeness and binding of inclusion paths
<code>Stego.extract_embed, dist_setLSB_le_one, join_chunks</code>	the embedding round-trips, perturbs a byte by at most one bit, and reassembles
<code>SecretSharing.existsUnique_missing_share</code>	any 24–1 shares leave every candidate key equally likely
<code>Paillier.enc_prod, msg_unique</code>	shares aggregate without being opened

5.7 Hiding, in theory and in practice

The perfect-hiding statement above is exactly the right one for this setting: it says that the published entry is a function of the statement and of the randomness alone, in the sense that for *any* other digest there is a unique randomness making an issuer who holds a different proof term publish the identical entry. Nothing about the proof term can therefore be read off the entry.

Independently of the theorem, a leak test samples 64-byte fragments of every withheld proof term and searches for them across everything published: the public artifacts, every file of every bundle, and the payload decoded back out of the images. Fragments that are copies of a published statement are counted separately, since statements are published on purpose; no other fragment occurs anywhere, and no proof term occurs in full.

5.8 Trust boundary, stated honestly

Checking the published objects establishes: that the certificate and the ledger are signed by the published key; that the records hash to the published roots; that each published statement matches its digest and carries an accepting transcript bound to it; that the transcripts are accepting for the published commitments; and, for a bundle, that every image matches its signed provenance and the deck reassembles to the payload it claims.

The openings themselves — the digests, and the randomness of the commitments — are the issuer’s private state, as is the kernel witness from which the digests are computed. Neither is part of what a bundle publishes.

Checking the published objects does *not* establish that the Lean library compiles on your machine. There is no circuit for the type-checker here, so this is a proof of *possession of kernel data whose acceptance the issuer certifies*, not a succinct argument that a type-checker would accept it. A reader who does not want to trust the extraction run should rebuild the library and re-run the extraction, which is deterministic and reproduces the digests; the certificate lists exactly which declarations, modules and axioms to expect. We claim the statement above and nothing more.

6 The proof object, in this document

6.1 How to check what follows

Let p, q, g, h be the group parameters printed in §6.2. Write $H(\text{dom}, x_1, \dots, x_n)$ for SHA-256 of the concatenation of the 4-byte big-endian length of the ASCII domain string, the domain string, and, for each x_i , its 8-byte big-endian length followed by x_i . Then, for a published entry with statement S , name N , commitment C , first message t and responses z, w :

1. the statement digest is SHA-256(S), over the UTF-8 bytes of the statement as published in the machine-readable ledger. What is typeset here is a faithful rendering of those bytes with the few characters that are special to $\text{\TeX}$ escaped, so recompute a digest from the published ledger file — which ships in every bundle — rather than from text lifted out of this PDF;
2. the Fiat–Shamir context is $\text{ctx} = H(\text{"zkp-statement-ledger"}, \text{ctx}_0, N, \text{SHA-256}(S))$, where ctx_0 is the ledger context `ff2612e3d07183acf0da8e3f9359431084a3f6faf8d75a7c7254b38b07622d51`;
3. the challenge is $c = H(\text{"schnorr-challenge"}, p, g, h, C, t, \text{ctx}) \bmod q$, each of p, g, h, C, t encoded as 256 big-endian bytes;
4. the transcript is accepting when $1 < C, t < p$, $C^q = t^q = 1 \bmod p$ and $g^z h^w = t C^c \bmod p$.

The Merkle leaf of an entry is $H(\text{"merkle-leaf"}, R)$ where R is the canonical JSON (sorted keys, no spaces) of the record pairing name, kind, module, the two statement digests, the commitment, the axiom list and the replay verdict; internal nodes are $H(\text{"merkle-node"}, \text{left}, \text{right})$, an odd node on a level being promoted unchanged. Those leaves fold to the ledger root `5a6125712e3e7c4a499dd11ad876f34f760c7eae34a4029325442de0cd27eb3c`, which is signed with Ed25519 under the issuer key `6f996285844038e962d600147c4faa74f6b7eb30b490865a72433dea415322ae`.

6.2 Group parameters, aggregate commitment, roots and signatures

Group. quadratic residues of the RFC 3526 2048-bit MODP group.

p

ffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74
020bbea63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f1437
4fe1356d6d51c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7ed
ee386bfb5a899fa5ae9f24117c4b1fe649286651ece45b3dc2007cb8a163bf05
98da48361c55d39a69163fa8fd24cf5f83655d23dca3ad961c62f356208552bb
9ed529077096966d670c354e4abc9804f1746c08ca18217c32905e462e36ce3b
e39e772c180e86039b2783a2ec07a28fb5c55df06f4c52c9de2bcbf695581718
3995497cea956ae515d2261898fa051015728e5a8aaca68ffffffffffff

q

fffffffffffffe487ed5110b4611a62633145c06e0e68948127044533e63a
0105df531d89cd9128a5043cc71a026ef7ca8cd9e69d218d98158536f92f8a1b
a7f09ab6b6a8e122f242dabb312f3f637a262174d31bf6b585ffae5b7a035bf6
f71c35fdad44cfd2d74f9208be258ff324943328f6722d9ee1003e5c50b1df82
cc6d241b0e2ae9cd348b1fd47e9267afc1b2ae91ee51d6cb0e3179ab1042a95d
cf6a9483b84b4b36b3861aa7255e4c0278ba3604650c10be19482f23171b671d
f1cf3b960c074301cd93c1d17603d147dae2aef837a62964ef15e5fb4aac0b8c
1ccaa4be754ab5728ae9130c4c7d02880ab9472d455655347ffffffffffff

g

4

h

95ebbf1e2d5dc0244665e09c12a5b39adf6eb9abed96aa919ee8b4e58609981a
b09cbe83b82c3c12d23658532d3364357b2c650d0a9abba587d2e6490649136e
b50191472598011527ec20aa3ce3927cb2665bada72250bf4df1979e5d0a1626
f2e9ed7b603b4578eca9d7977010f7c8f74f6b981176b80378a5eed57fdc5bc1
7e48adc937c0f992870730c1aa2246d8e872c41ccd6336dda62c0eeefed1e1d
0466359a245e5fddf361b08250cb218f88a02d6d0f67501bebea3adf80a81665
ca7d5c7ef2a9739e4d32de5e7de846932120208c2718d3e696435e6e03444f63
b56519e0d1052f5c1300ed189d70da2c209882f8dbf2bdd14b676e11138944a4

Aggregate commitment and its proof of knowledge.

C

9a7b9d6f4ac5d84c4ab790e86d250ab517029bcf0b669ba216cad20c440c863c
2f3475ba168e7394f2b3621f2d4caf2f34f1997e0cc9d7c485e4ebe730c0ca35
e69ae126da7a8902b649a1bd3a6b8cf3c533338518c9b3857b4ad42294d81cd7
877728397d1e3c6f318290f70f7dc161e8ef0e08a39fa91dfb185d4cd7a1c5ec
02d03c7b101f16346598c1c99b713d35617107a4b1d080998095ccf8d5b2864e
9bfb3be3ee2c8af3e674f7633a9f53af85f782f334bf5c301dcc5d1572ad3c7
9f5ae888a1185c1796dc7d5700f8b261b82f121d27fee17c5af8d39d14749c1c
c9857e864114f94dc789676f97b05f4e7d5cb11abc61ed553fadcd44a7de2f5d

t

2d1e22579e7976521625b6efc09673420bd01bd9331325088c738ea88c4f521e
d3eb13d1cc0e39b18d40a91bf64cf1c7ad0b4eaf3ab743edaee7315266d5f550
eaf7a017c396e96353e88cac81591a1e2e6e9b2a43f1781118592590f5e66ef6
18157d997d287baf84f70baab19ae7d2e08d2a3c4b5322a8f88afcde23e69a4d
61716c3094597a2e15ee8b03372947d9b09483fc70ef5d4e5a02fb4aacca195e
79f06270be6d9f4090e39b32444abb78896390aaf2b672c748a40b9bb4662ca7
a0a237c28ade26a18b5b00b01f4e72f3dde292325fc889f4719c77a13dc00471
8c8944f07fc0bbce63ad5917b0a3aad16cc5792b399a115d0202e9c87d590b7e

z

4a82b144a003f5bf5417ca548ff95e2bafd46af25a76abf88582c3b143e2212d
860289095274bc68973c8c5ada9d0490b8b4a3caf047197122f468710a3ff8c4
817205c9eda07df097582066c378ad2ef1458b457a57f807a3530fc2d57dacd9
388a5ff478ac20eb911448e5559d25d51ff27f5cbf7f171bfbf4547e7ad88ef9
e314b7bc999858b094f59c617976610adb89c508b79a409ca5afdc02bc4ab253
a0ed9ae01e5c3e99e9f04ff66d3bffe5e1145fb3f6bc8a78f3a944f9d56e30a4
cfd407714f0b603fffd36b09a3a08da243b30e5f9f8ac36b117ec09b273b0c81
7fed84772a66d34df80606d2207afa6b43ebe6c83ba1d3c94d13f19ca2724350

w

98d90c3eb719dfa35954836c320f32dbf81782da3fb215d5c2c49c0b1dad588a
1d82f2c5216a539f8353ba3b0a39f1fe7ac4a4d785b61a5660754910de7f80be
57e4ef30c618e428f4bc364353a47fc41fcfd435f1356f8869df553c950cbdd1
e096541ee0db58e4f2912bf1e0d5238db5d255fc971fa6178732adad8017901b
0f66cbbaff6269aee3d202b35ca1173b11259583a9ecc16646f6abb5fa1f9289
3c1605943ad19c2aa0b658bd94046e5280b6b68627545422cd3b265d7884738f
1871697e9679de3c58963b5e618c585e2d9a58ddfb3a82878413ef81049b99e4
13ae222341eb72f79d9b827daeb4addab35bb89b2a91d84bce0d85da77211d1

context

ff2612e3d07183acf0da8e3f9359431084a3f6faf8d75a7c7254b38b07622d51
616767726567617465

Roots, signatures and keys.

```

kernel Merkle root
d88f86c62c25bef0ab20652cc7a94a296b5c0f073c709cb24e8e21e94593b800
ledger Merkle root
5d6125712e3e7c4a499dd11ad876f34f760c7eae34a4029325442de0cd27eb3c
certificate signature
6959cc27fc8616bb75c4d725452839bcb006465b4b059944488b78020df3b54
9a687c84d48e28a18992f965cdf13b75498fe645a1a7a79a7d52829aff4d9702
ledger signature
1648bac5ce87a26f9dcb638c4a14716907b3b82a91f1a673bf6757f9c508f7a5
1e9896361cbc9fac169c10f58fa911c63d5143638f0d040197dfec396043840f
Ed25519 public key
6f996285844038e962d600147c4faa74f6b7eb30b490865a72433dea415322ae
Paillier modulus n
c2e70e1cedb2d2d264d2511c38d9690d0058db5d29920f69721f57561593da49
7f03425674f13544390338688440269d926ce0a57cca6a97d8f5896580de4acc
b3300d6e8569bb33bd90414e736f1347d272911830faaa124a1c8a093d6a2336
41ae58c51b5e53e01e8b18b81abf6c8bd12df75bc62eaf44098be9a5a45e91e8
3efe4c06a893d8f0eff7e26272715571e691a48f19d68ce8caef313329d8548c
025e2514086c8e669cd00fe61d07d8c756adb4d544759e1b17033e46a5dc896
69f9373fd9f333bae751c4aed36c9ec7b49a3e900c031f66e8a9a5aa34bb4d41
06fd0ac362c467f471e6416620ab4c102c9c1ead0d644a503e356d6f8712ab5d

```

6.3 Principal statements, in clear

Each block below is the Lean type of a declaration, printed with full names and universes — the statement in clear — followed by its provenance and the digest that binds it to its commitment. The proof term is not printed, here or anywhere; the number given is its size in bytes. The complete list of all 692 entries, each with its commitment and its full transcript, is the companion volume `ledger-appendix.pdf`.

FBundlePoint.gaugeSystem_iff_exists_Q

Classification of framed F -bundles over a point (simple eigenvalues).

```

∀ {ℓ : Type u_1} [inst : Field.{u_1} ℓ] {m : Nat} [CharZero.{u_1} ℓ] (ξ : Fin m → ℓ),
  Function.Injective.{1, u_1 + 1} ξ →
    ∀ (G K' G' : Matrix.{0, 0, u_1} (Fin m) (Fin m) ℓ),
      Iff
        (∃ P,
          And (IsUnit.{u_1} (Matrix.det.{u_1, 0} (P 0)))
              (FBundlePoint.GaugeSystem.{u_1} (Matrix.diagonal.{u_1, 0} ξ) G K' G' P))
        (∃ Q,
          And (IsUnit.{u_1} (Matrix.det.{u_1, 0} Q))
              (And
                (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (Matrix.diagonal.{u_1, 0} ξ) Q)
                  (HMul.hMul.{u_1, u_1, u_1} Q K'))
                (∀ (i : Fin m),
                  Eq.{u_1 + 1} (G i i)
                    (HMul.hMul.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q G')
                      (Inv.inv.{u_1} Q) i
                    i))))

```

source FBundlePoint, line 368 • withheld proof term: 125515 bytes • SHA-256 of the statement 0b8c37729f7e963d226b60eb4d0eca2ef0ccf448404d7e58aade853e8f8053b9

FBundlePoint.isGaugeEquiv_iff_exists_Q*The same classification for matrices over $\mathbb{k}[[u]]$.*

```

∀ {m : Nat} {F : Type u_2} [inst : Field.{u_2} F] [CharZero.{u_2} F] (ξ : Fin m → F),
  Function.Injective.{1, u_2 + 1} ξ →
    ∀ (G K' G' : Matrix.{0, 0, u_2} (Fin m) (Fin m) F),
      Iff
        (∃ Φ,
          And (IsUnit.{u_2} (Matrix.det.{u_2, 0} Φ))
              (FBundlePoint.IsGaugeEquiv.{u_2} (Matrix.diagonal.{u_2, 0} ξ) G K' G' Φ))
        (∃ Q,
          And (IsUnit.{u_2} (Matrix.det.{u_2, 0} Q))
              (And
                (Eq.{u_2 + 1} (HMul.hMul.{u_2, u_2, u_2} (Matrix.diagonal.{u_2, 0} ξ) Q)
                  (HMul.hMul.{u_2, u_2, u_2} Q K'))
                (∀ (i : Fin m),
                  Eq.{u_2 + 1} (G i i)
                    (HMul.hMul.{u_2, u_2, u_2} (HMul.hMul.{u_2, u_2, u_2} Q G')
                      (Inv.inv.{u_2} Q) i
                      i))))

```

source FBundlePointPowerSeries, line 151 • withheld proof term: 101 012 bytes • SHA-256 of the statement edcd50bece77b15d88e9594487f6728714e8ae2a893f95fc678591a5bfa9ee17

FBundlePoint.isBlockDiagOn_of_commutator_of_disjoint_spectrum*Disjoint spectra force a matrix commuting into a block decomposition to split.*

```

∀ {κ : Type u_1} [inst : Field.{u_1} κ] [IsAlgClosed.{u_1} κ] {ι : Type u_2} {κ : Type u_3}
  [inst_2 : Fintype.{u_2} ι] [inst_3 : DecidableEq.{u_2 + 1} ι] [inst_4 : DecidableEq.{u_3
    + 1} κ]
  {part : ι → κ} {U X : Matrix.{u_2, u_2, u_1} ι ι κ},
  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
    (∀ (k l : κ),
      Ne.{u_3 + 1} k l →
        ∀ (x : κ),
          Membership.mem.{u_1, u_1}
            (spectrum.{u_1, max u_1 u_2} κ (FBundlePoint.blockOf.{u_1, u_2, u_3} part U
              k k))
            x →
              x ∉ spectrum.{u_1, max u_1 u_2} κ (FBundlePoint.blockOf.{u_1, u_2, u_3} part
                U l l)) →
          FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
            (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
              (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)
              (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U)) →
          FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part X

```

source BlockPartition, line 297 • withheld proof term: 17 695 bytes • SHA-256 of the statement 5854d62da921112d6f1e1dd9d4328b9a351c59a16ab0983e3eaa78affb5a5235

FBundlePoint.existsUnique_commutator_isBlockOffDiagOn*The adjoint action is a bijection of the off-diagonal part onto itself.*

```

∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{U : Matrix.{u_2, u_2, u_1} ι ι κ},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part U →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part U l l)))
    →
  ∀ {C : Matrix.{u_2, u_2, u_1} ι ι κ},
  FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part C →
  ExistsUnique.{max (u_1 + 1) (u_2 + 1)} fun X =>
    And (FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part X)
      (Eq.{max (u_1 + 1) (u_2 + 1)}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} U X)
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} X U)))
    C)

```

source BlockPartition, line 314 • withheld proof term: 833 341 bytes • SHA-256 of the statement
 243c5c213bdb3e4fa91467ffb72abf8fa655bcf6e4edbf5158c13ffff8a84f79

FBundlePoint.exists_isGaugeEquivSeries_isBlockDiagOn*Spectral decomposition of a connection on the formal disk.*

```

∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
[inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
{K : Matrix.{u_2, u_2, u_1} ι ι κ},
FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
      IsCoprime.{u_1}
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
        (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l l)))
    →
  ∀ (A : Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} κ)),
  ∃ Φ A',
  And (Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ) 1)
    (And (IsUnit.{u_1} (Matrix.det.{u_1, u_2} Φ))
      (And
        (∀ (n : Nat),
          FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
            (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1) Φ))
        (And
          (∀ (n : Nat),
            FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
              (FBundlePoint.psCoeffM.{u_1, u_2} n A'))
            (FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ A')))))

```

source SpectralPowerSeries, line 191 • withheld proof term: 153 734 bytes • SHA-256 of the statement

FBundlePoint.isGaugeEquivSeries_unique*Uniqueness of that decomposition under the normalization.*

```

∀ {κ : Type u_1} [inst : Field.{u_1} κ] {ι : Type u_2} {κ : Type u_3} [inst_1 :
  Fintype.{u_2} ι]
  [inst_2 : DecidableEq.{u_2 + 1} ι] [inst_3 : DecidableEq.{u_3 + 1} κ] {part : ι → κ}
  {K : Matrix.{u_2, u_2, u_1} ι ι κ},
  FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part K →
  (∀ (k l : κ),
    Ne.{u_3 + 1} k l →
    IsCoprime.{u_1}
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K k k))
      (Matrix.charpoly.{u_1, u_2} (FBundlePoint.blockOf.{u_1, u_2, u_3} part K l l)))
    →
  ∀ {A Φ1 A1 Φ2 A2 : Matrix.{u_2, u_2, u_1} ι ι (PowerSeries.{u_1} κ)},
  Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ1) 1 →
  (∀ (n : Nat),
    FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
      (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1) Φ1)) →
  (∀ (n : Nat),
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
      (FBundlePoint.psCoeffM.{u_1, u_2} n A1)) →
  FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ1 A1 →
  Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.psCoeffM.{u_1, u_2} 0 Φ2) 1 →
  (∀ (n : Nat),
    FBundlePoint.IsBlockOffDiagOn.{u_1, u_2, u_3} part
      (FBundlePoint.psCoeffM.{u_1, u_2} (HAdd.hAdd.{0, 0, 0} n 1) Φ2)) →
  (∀ (n : Nat),
    FBundlePoint.IsBlockDiagOn.{u_1, u_2, u_3} part
      (FBundlePoint.psCoeffM.{u_1, u_2} n A2)) →
  FBundlePoint.IsGaugeEquivSeries.{u_1, u_2} K A Φ2 A2 →
  And (Eq.{max (u_1 + 1) (u_2 + 1)} Φ1 Φ2)
    (Eq.{max (u_1 + 1) (u_2 + 1)} A1 A2)

```

source SpectralPowerSeries, line 212 • withheld proof term: 40 875 bytes • SHA-256 of the statement
 22c389691bf534a58ddbef370d5125209189935f6ea4657922b9fc3e885876b8

FBundlePoint.mv_unique_of_mvPderiv_bracket

The t-direction matrix lemma over arbitrarily many formal variables.

```
∀ {ℓ : Type u_1} [inst : Field.{u_1} ℓ] {σ : Type u_2} {m : Nat} [CharZero.{u_1} ℓ]
  (T : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ ℓ))
  (X Y : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ ℓ)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (X k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T i) (X (HAdd.hAdd.{0, 0,
            0} k 1))))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k 1))
            (T i)))))) →
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (Y k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (T i)
            (Y (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Y (HAdd.hAdd.{0, 0, 0} k
            1))
            (T i)))))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k))
      (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Y k))) →
    Eq.{max (u_1 + 1) (u_2 + 1)} X Y
```

source MatrixMultiDirection, line 262 • withheld proof term: 912001 bytes • SHA-256 of the statement
5f72ab50d60c1e2a000f464eb8b28628f25ec762eca6df893339a3b9ef034056

FBundlePoint.mv_unique_of_mvEuler_bracket

The logarithmic matrix lemma, under the nilpotency condition.

```
∀ {ℓ : Type u_1} [inst : Field.{u_1} ℓ] {σ : Type u_2} {m : Nat} [CharZero.{u_1} ℓ]
  (Q : σ → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ ℓ)),
  (∀ (i : σ),
    IsNilpotent.{u_1} (FBundlePoint.adMat.{u_1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Q
      i)))) →
  ∀ (X Y : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ
    ℓ)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvEuler.{u_1, u_2} i (X k))
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Q i)
          (X (HAdd.hAdd.{0, 0, 0} k 1)))
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k
          1))
            (Q i)))) →
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvEuler.{u_1, u_2} i (Y k))
      (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Q i)
          (Y (HAdd.hAdd.{0, 0, 0} k 1)))
        (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (Y (HAdd.hAdd.{0, 0, 0} k
          1))
            (Q i)))) →
  (∀ (k : Nat),
    Eq.{u_1 + 1} (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X k))
      (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (Y k))) →
  Eq.{max (u_1 + 1) (u_2 + 1)} X Y
```

source MatrixMultiDirection, line 350 • withheld proof term: 740 221 bytes • SHA-256 of the statement
8d75d12ca8845606effd5c00219e0f6dc502c36814a90d7c3ad08034ead46b85

FBundlePoint.existsUnique_derivPS_bracket

Well-posedness of the formal Cauchy problem in one direction.

```
∀ {ℓ : Type u_1} [inst : Field.{u_1} ℓ] [CharZero.{u_1} ℓ] {m : Nat}
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} ℓ))
  (C : Nat → Matrix.{0, 0, u_1} (Fin m) (Fin m) ℓ),
  ExistsUnique.{u_1 + 1} fun X =>
    And
      (∀ (k : Nat),
        Eq.{u_1 + 1} (FBundlePoint.derivPS.{u_1} (X k))
          (Neg.neg.{u_1}
            (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} T (X (HAdd.hAdd.{0, 0, 0}
              k 1)))
              (HMul.hMul.{u_1, u_1, u_1} (X (HAdd.hAdd.{0, 0, 0} k 1)) T))))
        (∀ (k : Nat), Eq.{u_1 + 1} (FBundlePoint.psCoeff.{u_1} 0 (X k)) (C k))
```

source TDirectionExistence, line 95 • withheld proof term: 75 553 bytes • SHA-256 of the statement
65cffa963e101fcb95cbfdd0550a17cef2b9b07c0bfb8d1bc7bf9a9e19478e9

FBundlePoint.commute_bracket_of_mvPderiv_bracket

The integrability obstruction in several directions.

```
∀ {ℓ : Type u_1} [inst : Field.{u_1} ℓ] {σ : Type u_2} {m : Nat}
  (T : σ → Matrix.{0, 0, u_1} (Fin m) (Fin m) ℓ)
  (X : Nat → Matrix.{0, 0, max u_1 u_2} (Fin m) (Fin m) (MvPowerSeries.{u_2, u_1} σ ℓ)),
  (∀ (i : σ) (k : Nat),
    Eq.{max (u_1 + 1) (u_2 + 1)} (FBundlePoint.mvPderiv.{u_1, u_2} i (X k))
      (Neg.neg.{max u_1 u_2}
        (HSub.hSub.{max u_1 u_2, max u_1 u_2, max u_1 u_2}
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (FBundlePoint.cstMv.{u_1,
            u_2} (T i))
            (X (HAdd.hAdd.{0, 0, 0} k 1)))
          (HMul.hMul.{max u_1 u_2, max u_1 u_2, max u_1 u_2} (X (HAdd.hAdd.{0, 0, 0} k 1))
            (FBundlePoint.cstMv.{u_1, u_2} (T i)))))) →
    ∀ (i j : σ) (k : Nat),
      Commute.{u_1}
        (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (T i) (T j))
          (HMul.hMul.{u_1, u_1, u_1} (T j) (T i)))
        (FBundlePoint.mvPsCoeff.{u_1, u_2} 0 (X (HAdd.hAdd.{0, 0, 0} k 2)))
```

source MvIntegrability, line 68 • withheld proof term: 4168212 bytes • SHA-256 of the statement
44bbcc3e240447aaff127c9a23c45492903f801e988828756dd46949bbc81bf0

FBundlePoint.isNilpotent_adMat_of_framing

A strong framing forces the nilpotency condition.

```
∀ {ℓ : Type u_1} [inst : Field.{u_1} ℓ] {m : Nat} [CharZero.{u_1} ℓ]
  (K G Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) ℓ),
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q K) (HMul.hMul.{u_1, u_1, u_1}
      K Q))
    0 →
  Eq.{u_1 + 1}
    (HSub.hSub.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} Q G) (HMul.hMul.{u_1, u_1,
      u_1} G Q))
    (Neg.neg.{u_1} Q) →
    And (IsNilpotent.{u_1} Q) (IsNilpotent.{u_1} (FBundlePoint.adMat.{u_1} Q))
```

source Nilpotency, line 123 • withheld proof term: 54821 bytes • SHA-256 of the statement 9e613558
8f2af1df8190ae3b307d3c44ddcc0386eebf8a3a5b949994705989f2

FBundlePoint.existsUnique_isFramedT

Existence and uniqueness of the framing gauge transformation in a t-direction.

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (T : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  ExistsUnique.{u_1 + 1} fun PTt =>
    And (Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} 0 PTt.1) 1)
      (FBundlePoint.IsFramedT.{u_1} T PTt.1 PTt.2)
```

source TFramingSeries, line 177 • withheld proof term: 207174 bytes • SHA-256 of the statement
f602c4a06a1f754e26b273535cc054803334058b3c0d19b35d4499d478abb1a5

FBundlePoint.existsUnique_isFramedQ

The same in a logarithmic direction, under the nilpotency condition.

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (A : Matrix.{0, 0, u_1} (Fin m) (Fin m) R),
  IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} A) →
  ∀ (Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
  Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} 0 Q) (FBundlePoint.cst.{u_1} A) →
  ExistsUnique.{u_1 + 1} fun PQt =>
    And (Eq.{u_1 + 1} (FBundlePoint.tCoeffM.{u_1} 0 PQt.1) 1)
        (FBundlePoint.IsFramedQ.{u_1} Q PQt.1 PQt.2)
```

source QFramingSeries, line 90 • withheld proof term: 240 054 bytes • SHA-256 of the statement 8bf40f46
edb4e31e1e1e51bce481fa105c30437579f20e20bfafe7807e207408

FBundlePoint.eq_cst_gauge_of_curvature_eq_zero

One step of the extension of framing.

```
∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (d_2 :
    AddMonoidHom.{u_1, u_1}
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))),
  (∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (d_2 (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d_2 x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d_2 y)))) →
  (∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (FBundlePoint.uDerivT.{u_1} (d_2 x)) (d_2 (FBundlePoint.uDerivT.{u_1}
      x))) →
  ∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
    R))},
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
  Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
  ∀ {T' A' : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
    (PowerSeries.{u_1} R))}
    {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)},
  Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} FBundlePoint.uDerivT.{u_1} P Q T')
    (FBundlePoint.cst.{u_1} T) →
  Eq.{u_1 + 1} (d_2 (FBundlePoint.cst.{u_1} T))
    (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1}
      (FBundlePoint.cst.{u_1} B)) →
  Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} FBundlePoint.uDerivT.{u_1} d_2
    T' A')
    0 →
  (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
    FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1)
      (FBundlePoint.gauge.{u_1} d_2 P Q A')) →
  Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} d_2 P Q A')
    (FBundlePoint.cst.{u_1}
      (FBundlePoint.psCoeff.{u_1} 0 (FBundlePoint.gauge.{u_1} d_2 P Q
        A'))))
```

source FlatnessGauge, line 219 • withheld proof term: 53 143 bytes • SHA-256 of the statement 70ebce94
9ff8f423e701f1fd18722a5300e21445c6ff699537383c77016bab54

FBundlePoint.eq_cst_gauge_of_curvature_eq_zero_q

The same step for a logarithmic direction.

```

∀ {R : Type u_1} [inst : CommRing.{u_1} R] {m : Nat} [Algebra.{0, u_1} Rat R]
  (d₂ :
    AddMonoidHom.{u_1, u_1}
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R)))
      (Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))))
  (∀ (x y : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
  (∀ (x : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1} R))),
    Eq.{u_1 + 1} (FBundlePoint.uEulerQ.{u_1} (d₂ x)) (d₂ (FBundlePoint.uEulerQ.{u_1}
      x))) →
  ∀ {P Q : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} (PowerSeries.{u_1}
    R))},
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
    ∀ {T' A' : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1}
      (PowerSeries.{u_1} R))},
      {T B : Matrix.{0, 0, u_1} (Fin m) (Fin m) (PowerSeries.{u_1} R)},
      IsNilpotent.{u_1} (FBundlePoint.adMatR.{u_1} (FBundlePoint.psCoeffR.{u_1} 0
        T)) →
      Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} FBundlePoint.uEulerQ.{u_1} P Q T')
        (FBundlePoint.cst.{u_1} T) →
      Eq.{u_1 + 1} (d₂ (FBundlePoint.cst.{u_1} T))
        (HSMul.hSMul.{u_1, u_1, u_1} PowerSeries.X.{u_1}
          (FBundlePoint.cst.{u_1} B)) →
      Eq.{u_1 + 1} (FBundlePoint.curvature.{u_1} FBundlePoint.uEulerQ.{u_1}
        d₂ T' A')
        0 →
      (FBundlePoint.IsFramedDirAt.{u_1} fun k =>
        FBundlePoint.psCoeff.{u_1} (HAdd.hAdd.{0, 0, 0} k 1)
          (FBundlePoint.gauge.{u_1} d₂ P Q A')) →
      Eq.{u_1 + 1} (FBundlePoint.gauge.{u_1} d₂ P Q A')
        (FBundlePoint.cst.{u_1}
          (FBundlePoint.psCoeff.{u_1} 0 (FBundlePoint.gauge.{u_1} d₂ P Q
            A'))))

```

source FlatnessGauge, line 261 • withheld proof term: 59 889 bytes • SHA-256 of the statement f504593f5d0f21133c04b6e8a20635649b6c881a1422bdcc7f7a8e7cf9247886

FBundlePoint.curvature_gauge

Gauge covariance of the curvature.

```
∀ {A : Type u_1} [inst : Ring.{u_1} A] (d₁ d₂ : AddMonoidHom.{u_1, u_1} A A),
  (∀ (x y : A),
    Eq.{u_1 + 1} (d₁ (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₁ x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d₁ y)))) →
  (∀ (x y : A),
    Eq.{u_1 + 1} (d₂ (HMul.hMul.{u_1, u_1, u_1} x y))
      (HAdd.hAdd.{u_1, u_1, u_1} (HMul.hMul.{u_1, u_1, u_1} (d₂ x) y)
        (HMul.hMul.{u_1, u_1, u_1} x (d₂ y)))) →
  (∀ (x : A), Eq.{u_1 + 1} (d₁ (d₂ x)) (d₂ (d₁ x))) →
  ∀ {p q : A},
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} p q) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} q p) 1 →
    ∀ (X₁ X₂ : A),
      Eq.{u_1 + 1}
        (FBundlePoint.curvature.{u_1} d₁ d₂ (FBundlePoint.gauge.{u_1} d₁ p q X₁)
          (FBundlePoint.gauge.{u_1} d₂ p q X₂))
        (HMul.hMul.{u_1, u_1, u_1}
          (HMul.hMul.{u_1, u_1, u_1} q (FBundlePoint.curvature.{u_1} d₁ d₂ X₁
            X₂)) p)
```

source GaugeCurvature, line 75 • withheld proof term: 3 781 778 bytes • SHA-256 of the statement
b6084a09f9ce870c3c948a89e268ad2b817b14f0fa58432f758eba198bedccac

FBundlePoint.diag_conj_diagonal_mem_of_companionK

The corrected vanishing-diagonal lemma: the entries vanish modulo the Chern classes.

```
∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (I : Ideal.{u_1} A)
  (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
  (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)), Membership.mem.{u_1, u_1} I (b k)) →
  ∀ (P Q : Matrix.{0, 0, u_1} (Fin (HAdd.hAdd.{0, 0, 0} m 1)) (Fin (HAdd.hAdd.{0, 0, 0} m
    1)) A),
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1 →
    Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1 →
    ∀ (lam : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
      Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} (FBundlePoint.companionK.{u_1} b) P)
        (HMul.hMul.{u_1, u_1, u_1} P (Matrix.diagonal.{u_1, 0} lam)) →
      ∀ (mu : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
        Membership.mem.{u_1, u_1} I (∑ j, mu j) →
        ∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          Membership.mem.{u_1, u_1} I
            (HMul.hMul.{u_1, u_1, u_1}
              (HMul.hMul.{u_1, u_1, u_1} Q (Matrix.diagonal.{u_1, 0} mu)) P i i)
```

source ProjectiveBundle, line 193 • withheld proof term: 2 821 122 bytes • SHA-256 of the statement
5e503ef060ceff1722eae4555b35c6a873e4fff46d94773c80bf2cef3affb72b

FBundlePoint.dualNumber_diag_conj_diagonal_ne_zero_of_any
A counterexample to the lemma as originally stated.

```

∀ (P Q : Matrix.{0, 0, 0} (Fin 2) (Fin 2) (DualNumber.{0} Rat)),
  Eq.{1} (HMul.hMul.{0, 0, 0} Q P) 1 →
    Eq.{1} (HMul.hMul.{0, 0, 0} P Q) 1 →
      Eq.{1}
        (HMul.hMul.{0, 0, 0}
          (FBundlePoint.companionK.{0} (FBundlePoint.cexB.{0} DualNumber.eps.{0})) P
            (HMul.hMul.{0, 0, 0} P
              (Matrix.diagonal.{0, 0}
                (FBundlePoint.cexLam.{0} DualNumber.eps.{0} FBundlePoint.qHalf)))) →
        Ne.{1}
          (HMul.hMul.{0, 0, 0}
            (HMul.hMul.{0, 0, 0} Q (Matrix.diagonal.{0, 0} FBundlePoint.cexMu.{0})) P 0 0)
          0

```

source ProjectiveBundle, line 468 • withheld proof term: 39472 bytes • SHA-256 of the statement
 ae1f1bd4ac9b319b613a0eb162da42e4d8ec289da021a8102d0b3e3375900858

FBundlePoint.exists_diagonalization_companionK
Diagonalization of the companion matrix by Hensel lifting.

```

∀ {A : Type u_1} [inst : CommRing.{u_1} A] {m : Nat} (b : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A)
  (I : Ideal.{u_1} A) {N : Nat},
  Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} I N) Bot.bot.{u_1} →
    (∀ (k : Fin (HAdd.hAdd.{0, 0, 0} m 1)), Membership.mem.{u_1, u_1} I (b k)) →
      ∀ (zeta : Fin (HAdd.hAdd.{0, 0, 0} m 1) → A),
        (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
          Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} (zeta i) (HAdd.hAdd.{0, 0, 0} m 1)) 1) →
            (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
              IsUnit.{u_1}
                (HMul.hMul.{u_1, u_1, u_1} (HAdd.hAdd.{u_1, u_1, u_1} (↑m) 1)
                  (HPow.hPow.{u_1, 0, u_1} (zeta i) m))) →
                (∀ (i j : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
                  Ne.{1} i j → IsUnit.{u_1} (HSub.hSub.{u_1, u_1, u_1} (zeta i) (zeta j))) →
                  ∃ lam P Q,
                    And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} Q P) 1)
                      (And (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1} P Q) 1)
                        (And
                          (Eq.{u_1 + 1} (HMul.hMul.{u_1, u_1, u_1}
                            (FBundlePoint.companionK.{u_1} b) P)
                            (HMul.hMul.{u_1, u_1, u_1} P (Matrix.diagonal.{u_1, 0} lam)))
                          (∀ (i : Fin (HAdd.hAdd.{0, 0, 0} m 1)),
                            Membership.mem.{u_1, u_1} I (HSub.hSub.{u_1, u_1, u_1} (lam i)
                              (zeta i))))))

```

source CompanionDiagonalization, line 424 • withheld proof term: 475781 bytes • SHA-256 of the
 statement ead5cd9235a72130b117ddaf024f2023755aaee4da6cb563e3f8b8bd24ce5b1e

ZKP.Ledger.exists_unique_view_match

The published ledger entry is perfectly hiding for the proof term.

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} [inst_1 : Fact
  (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) (s : H) {y : ZMod q},
  (Function.Injective.{1, u_1 + 1} fun a => ZKP.Sigma.gpow.{u_1} g a) →
  Eq.{u_1 + 1} h (ZKP.Sigma.gpow.{u_1} g y) →
  Ne.{1} y 0 →
  ∀ (m_1 r_1 m_2 a b c : ZMod q),
  ExistsUnique.{1} fun r_2 =>
    Eq.{max (u_1 + 1) (u_2 + 1)}
      (ZKP.Ledger.honestView.{u_1, u_2} g h s m_2 r_2
        (ZKP.Ledger.shift c m_1 r_1 m_2 r_2 (Prod.mk.{0, 0} a b)).1
        (ZKP.Ledger.shift c m_1 r_1 m_2 r_2 (Prod.mk.{0, 0} a b)).2 c)
      (ZKP.Ledger.honestView.{u_1, u_2} g h s m_1 r_1 a b c)
```

source Crypto.Ledger, line 125 • withheld proof term: 13073 bytes • SHA-256 of the statement 419d4e8a
cfaee874aefc3d43c433622e10d4df47e52583671b29eda6057039fb

ZKP.Ledger.extract_opening_of_two_views

Soundness: two accepting entries extract the digest of the proof term.

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {H : Type u_2} {q : Nat} [inst_1 : Fact
  (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h : G) {v_1 v_2 : ZKP.Ledger.View.{u_1, u_2} G H q},
  ZKP.Ledger.Valid.{u_1, u_2} g h v_1 →
  ZKP.Ledger.Valid.{u_1, u_2} g h v_2 →
  Eq.{u_1 + 1} (ZKP.Ledger.View.commitment.{u_1, u_2} v_1)
    (ZKP.Ledger.View.commitment.{u_1, u_2} v_2) →
  Eq.{u_1 + 1} (ZKP.Ledger.View.first.{u_1, u_2} v_1)
    (ZKP.Ledger.View.first.{u_1, u_2} v_2) →
  Ne.{1} (ZKP.Ledger.View.challenge.{u_1, u_2} v_1)
    (ZKP.Ledger.View.challenge.{u_1, u_2} v_2) →
  Eq.{u_1 + 1} (ZKP.Ledger.View.commitment.{u_1, u_2} v_1)
    (ZKP.Sigma.commit.{u_1} g h
      (HMul.hMul.{0, 0, 0}
        (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.respM.{u_1, u_2} v_1)
          (ZKP.Ledger.View.respM.{u_1, u_2} v_2))
        (Inv.inv.{0}
          (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.challenge.{u_1, u_2} v_1)
            (ZKP.Ledger.View.challenge.{u_1, u_2} v_2))))
        (HMul.hMul.{0, 0, 0}
          (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.respR.{u_1, u_2} v_1)
            (ZKP.Ledger.View.respR.{u_1, u_2} v_2))
          (Inv.inv.{0}
            (HSub.hSub.{0, 0, 0} (ZKP.Ledger.View.challenge.{u_1, u_2} v_1)
              (ZKP.Ledger.View.challenge.{u_1, u_2} v_2)))))))
```

source Crypto.Ledger, line 156 • withheld proof term: 6076 bytes • SHA-256 of the statement f447f078
cef9a8da763bfb3a8b8d2e6e6250287d597648a1605c46f9ddb01f2a

ZKP.Sigma.sigma_special_soundness

Special soundness of the Σ -protocol.

```
∀ {G : Type u_1} [inst : CommGroup.{u_1} G] {q : Nat} [inst_1 : Fact (Nat.Prime q)],
  (∀ (x : G), Eq.{u_1 + 1} (HPow.hPow.{u_1, 0, u_1} x q) 1) →
  ∀ (g h C t : G) {c_1 c_2 z_1 z_2 w_1 w_2 : ZMod q},
  Ne.{1} c_1 c_2 →
  ZKP.Sigma.Accepts.{u_1} g h C t c_1 z_1 w_1 →
  ZKP.Sigma.Accepts.{u_1} g h C t c_2 z_2 w_2 →
  Eq.{u_1 + 1} C
  (ZKP.Sigma.commit.{u_1} g h
    (HMul.hMul.{0, 0, 0} (HSub.hSub.{0, 0, 0} z_1 z_2)
      (Inv.inv.{0} (HSub.hSub.{0, 0, 0} c_1 c_2))))
  (HMul.hMul.{0, 0, 0} (HSub.hSub.{0, 0, 0} w_1 w_2)
    (Inv.inv.{0} (HSub.hSub.{0, 0, 0} c_1 c_2))))
```

source Crypto.Sigma, line 146 • withheld proof term: 158 544 bytes • SHA-256 of the statement 5998a93de8de3a768d9857ec47fd7104628b706ad8b35ec8305de5c9a61ff94e

7 Verification results

Everything reported in this document was re-run against the artifacts as published, on the same day they were generated.

- **The library.** lake build completes (2149 jobs); the repository harness reports 10 of 10 checks passing: no sorry or admit, no axiom declaration, no native_decide or @[implemented_by], every declaration's axiom closure inside Classical.choice, Quot.sound, propext, the vacuity examples, the executed recursion benchmark, and every module covered.
- **The ledger.** All five ledger checks pass: the Ed25519 signature; 692 statements published in clear, each matching its digest; 692 accepting zero-knowledge transcripts, each bound to its statement; the statements and commitments hashing to the published ledger root; and no proof term anywhere in the ledger, with 159 288 447 bytes withheld.
- **Freshness.** A fresh extraction from the compiled library reproduces the digests: all 18 freshness checks pass, including that every one of the 692 published commitments opens to the digest of the current withheld proof term (692/692 open), that the aggregate is the product of the individual commitments, and that no declaration shares a witness file with another.
- **The bundles.** All 15 end-to-end tests pass: the 24 bundles are present and verify; a bundle verifies from its images alone; a deck with one image missing does not decode and a single flipped pixel breaks the hidden chunk; all 24 bundles together recover the private key and any 24–1 do not; the bundles are pairwise distinct views; and a leaked image is traced back to its bundle.
- **Leakage.** Of 2756 sampled 64-byte fragments of withheld proof terms, 2731 are not part of any published statement, and none of those occurs anywhere in the published corpus; no proof term occurs in full; every statement of the library is published, in clear, and none is empty. The largest bundle is 6.3 MB against 159.3 MB of withheld proof data.

8 What was found, and what remains

Formalizing changed the mathematics in one place and sharpened it in several others. The vanishing-diagonal lemma of the projective-bundle application is false as stated for a nontrivial

bundle (§3.8); the corrected statement, modulo the ideal of Chern classes, is what the application needs and is what is proved. The hypotheses of both matrix lemmas are shown to be necessary by explicit counterexamples, the one-variable and multivariate formulations are proved to agree, the existence half of the one-direction system is complemented by an explicit integrability obstruction in several directions, and the framing lemmas are proved in the coefficient-ring generality in which the paper actually applies them rather than only over a field.

What remains out of scope is the geometry: F-bundles over formal schemes and non-archimedean analytic spaces, and therefore the global decomposition and extension-of-framing theorems and the quantum-cohomology applications as statements about spaces. Their linear-algebraic inputs are here; the geometric infrastructure is not in Mathlib, and building it is a separate undertaking.

Reproducing

Building the library, running the harness, re-extracting the kernel data, rebuilding the bundles and re-verifying the published objects are all scripted in the repository; the repository's `README` lists the commands, and `zfp/PROTOCOL.md` states the protocol in the same terms as §5. This document and its companion volume are regenerated from the published artifacts by `tools/paper.py` followed by two LaTeX runs, so they cannot drift from the objects they describe.

A Index of published statements

Every declaration whose statement is published, with the SHA-256 of the published statement text. The statements themselves, the commitments and the transcripts are in the companion volume.

declaration	kind	SHA-256 of the statement
<i>BlockPartition</i>		
<code>FBundlePoint.IsBlockDiagOn</code>	def	931a62c08cb16bb139e02c785586b47934e4620a85ecfda7d748849249a1eef9
<code>FBundlePoint.IsBlockDiagOn.add</code>	thm	df90c9117cabf53acc498aa4b8ee8c461fc8dba0af804ca927784eae3466f27a
<code>FBundlePoint.IsBlockDiagOn.mul</code>	thm	1154fbabe41376c8df6574421a794cce1d1a0dbd5c6eafd82f794e8765532a07
<code>FBundlePoint.IsBlockDiagOn.neg</code>	thm	355be95fad87dbb313372526c8837268bd67d6a1f6d0a6a4f6ca5ef649460936
<code>FBundlePoint.IsBlockDiagOn.sub</code>	thm	58504b09707aa716559791f2b76996a8f56d1af2948747c1ed33d62564cdb730
<code>FBundlePoint.IsBlockDiagOn.sum</code>	thm	868fd4ea9ebf9b2e4e24311b777c0a502c07e52d4dafc33557e2f89b2cc18eb3
<code>FBundlePoint.IsBlockOffDiagOn</code>	def	931a62c08cb16bb139e02c785586b47934e4620a85ecfda7d748849249a1eef9
<code>FBundlePoint.IsBlockOffDiagOn.sub</code>	thm	8d0fd2340667fa9d86ac28c9d8315833e735dc6677a7ae4916b7b398446a5f68
<code>FBundlePoint.blockOf</code>	def	1f894ff1f2744ecd623f75c73037e1db74c5caa2f2f2bae97df70ff0cf5e9241
<code>FBundlePoint.blockOf.eq_1</code>	thm	34f484deaaf1ca2abc8ab431db1a01328816c65ea1700bce1caf1465ec4f3885
<code>FBundlePoint.blockOf_apply</code>	thm	2cd212cc8adad7b4686a5785e7c84b9dc5e421bfa08f8b1ed8bd4977de9a5468
<code>FBundlePoint.blockOf_commutator</code>	thm	0bc41aca2f7802777510b5d78e865543b87a964b9231be4677d6e4328dca5ecf
<code>FBundlePoint.blockOf_diagonal</code>	thm	cf9a32dda72de06d153c031ae86360c3df1b9bbf949ed2586e71e743ae5eb6f4

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.blockOf_mul_of_isBlockDiagOn_left</code>	thm	0793fa61e156c09a4e0553466074e0296f5cc97c81c359c314cd582a2a8aeebf
<code>FBundlePoint.blockOf_mul_of_isBlockDiagOn_right</code>	thm	364d1023a83e6cef061f2dda5c3a4f3ff2cebf7d43cdfd911ae3f8e20ed0d308
<code>FBundlePoint.eq_zero_of_isBlockDiagOn_of_isBlockOffDiagOn</code>	thm	28fa68a6d98354e80dba34879b2d3583cff53d442a6bf19ded4a06ff6be1b4b0
<code>FBundlePoint.existsUnique_commutator_isBlockOffDiagOn</code>	thm	243c5c213bdb3e4fa91467ffb72abf8fa655bcf6e4edbf5158c13ffff8a84f79
<code>FBundlePoint.exists_sub_commutator_isBlockDiagOn</code>	thm	ce0017aa4979f62ed47060f4b816e735af78dea4e2fd27977979a48de3424172
<code>FBundlePoint.isBlockDiagOn_diagonal</code>	thm	312758193c3793f9fa07d1fe238e14c21857269785b868c9eba0da993ecb4c8e
<code>FBundlePoint.isBlockDiagOn_iff_blockOf</code>	thm	a2402fbd9a40d35ce4fa57fb7959d79040574081f37700969b248037f52a37f
<code>FBundlePoint.isBlockDiagOn_of_commutator_isBlockDiagOn</code>	thm	5ba85481019d06a0363c9233d4edfc229a4dcb77162859042b4ad774b13c8b38
<code>FBundlePoint.isBlockDiagOn_of_commutator_of_disjoint_spectrum</code>	thm	5854d62da921112d6f1e1dd9d4328b9a351c59a16ab0983e3eaa78affb5a5235
<code>FBundlePoint.isBlockDiagOn_of_commute</code>	thm	cfa16ab5344c7876debd4df2fd73d0bc47d2e9b2afde465005c13f35b8ec68c6
<code>FBundlePoint.isBlockDiagOn_of_flatness</code>	thm	a1e831378a902bdcedf465651c5e8d7cf7598a8179a3dd29f43b288626da501e
<code>FBundlePoint.isBlockDiagOn_sub_offDiagPart</code>	thm	259cab76941bd3a306a9f2e3763cb50980919698cb0302fba10e1b421e647261
<code>FBundlePoint.isBlockDiagOn_zero</code>	thm	cc3f607f9df5063e1f44c33020c466690885a21d584bef5b5b541233d8ababd6
<code>FBundlePoint.isBlockOffDiagOn_commutator</code>	thm	4d11ceca1f75a2c446079a5f3fb4909b398ff24639c30e949fd116ba6a0b7f66
<code>FBundlePoint.isBlockOffDiagOn_iff_blockOf</code>	thm	ed8d9f1e56bafcbff2bd78770ca0860d68528364aaead8f250405af604f7226c
<code>FBundlePoint.isBlockOffDiagOn_offDiagPart</code>	thm	300c7df7f6b7456a8585538c59758edb44bcef7f87ccfe227f6d150eb4b8ef10
<code>FBundlePoint.offDiagPart</code>	def	c12134d08da9b3416c712058c67bed8206c71cd4e4639ab427bc29f59daff6ea
<code>FBundlePoint.offDiagPart.eq_1</code>	thm	b557058d1076de5162bc16ce6f13c0b3c468c7d8c605526f96a5d5aeb420d9e5
<code>FBundlePoint.sum_eq_sum_fibre</code>	thm	68f5a113938600c337f0b966ece464573d1ad7045b4c2f4769a8a45c2df63a3c
<i>CompanionDiagonalization</i>		
<code>FBundlePoint.castSucc_add_one</code>	thm	63fb27b1d297a716d1bd155209e17c88c63763dbb7b598c5e53c7f7f9e89b3f8
<code>FBundlePoint.companionK_mul_eigenMat</code>	thm	d88c8b87abb3a516d8244690f863e48a7f6e8a9c1ce9e6941b60811ca3a306ef
<code>FBundlePoint.companionPoly</code>	def	efc1a4ba26710cb409082664fdbe7e8c210e8699f50e624e26051cf511f153df
<code>FBundlePoint.companionPoly.eq_1</code>	thm	df8c10e5561f70eccc0347089a8aea0639018a7be27d5f0ea6e6cddb738f8021
<code>FBundlePoint.companionPoly_derivative_eval</code>	thm	bb0d8a0d14c68551b27d4c0d735eb1e576ae86807392303cd607acd73efdfb81
<code>FBundlePoint.companionPoly_eval</code>	thm	8f9791f44d53728fdf204bb926127f8d6159e3780e42d77caa6c52e4adcfc21b
<code>FBundlePoint.companionPoly_firstChern_eval</code>	thm	9706c9f7e118d04af8b1fc86167e37d8547c9a64c2403752a5334f267bd2bd1f
<code>FBundlePoint.companionPoly_firstChern_root</code>	thm	a789d4f1ee92ec2c6ddbe424aa25b3095199ba31ddd8fe975891fb56ae8a1b23
<code>FBundlePoint.companionPoly_moniac</code>	thm	75ebd916386dcc69dc74a065a605c8b83326faa7bdfa9fca9974f18b4328f281

declaration	kind	SHA-256 of the statement
FBundlePoint.diag_conj_diagonal_mem_of_n ilpotent	thm	f5daf0cf6ddf71bc64c12d480d0bbc1bab7cfc7c 5b60a2535bd8d47f9bc3bd95
FBundlePoint.eigenCol	def	c6194b0d6a2f8af7b07d5245231de57bd5bf073b a6e79252e8581a6963167a6d
FBundlePoint.eigenCol.eq_1	thm	62c25a26636307b5338df4c9c9685a01a4df3f5f 576cc7929435b24d443c702e
FBundlePoint.eigenCol_last	thm	29c596a802021cd56460bae8178d6cce76bd6fd3 fff52e4afdc3e3959499093b
FBundlePoint.eigenCol_of_lt	thm	ee09383508ead608d7d71fcc79c22c940e5fb921 2a86c802cb28f1db9e8ff6a9
FBundlePoint.eigenCol_zero	thm	46809071217f222ed4f3e39e9f162d9df61c4e54 af1bae1ee7326e0db3715c50
FBundlePoint.eigenMat	def	b218848a9f8c32c984dab7d0dcbbf4fc02b183ea 3da4c9587f972929d3e5442c
FBundlePoint.eigenMat.eq_1	thm	ee2932a2425aa9f7274cbe843c9420d33007e4be 323a4025d39c8f461ba9847b
FBundlePoint.eval_sub_mem_of_sub_mem	thm	f409011350a844e00b0f58cbfb6675315c8e85f4 0d4f9817a228229d9b440daf
FBundlePoint.exists_diagonalization_comp anionK	thm	ead5cd9235a72130b117ddaf024f2023755aaee4 da6cb563e3f8b8bd24ce5b1e
FBundlePoint.exists_diagonalization_firs tChern	thm	c8a1883caf2e2ef55d46dae6555111e9b48f5b0b 2091a3bc4445f1aac3d178d8
FBundlePoint.exists_diagonalizing_of_roots	thm	a9f928db04fdbel309774556f15630cedd731570 9b1555b4382316fa9cdd3910
FBundlePoint.exists_isRoot_of_pow_eq_bot	thm	682713485e41254513ace2a3963d1c120e31c86d 2658bb286c566957ed86b9e6
FBundlePoint.exists_roots_companionPoly	thm	5fa8b9e4558f8fd554e119b05bdc07dbad734b3e 41084f303ea346efb7483e0c
FBundlePoint.firstChernCol	def	53c8c75c7a8f299ee9b70368d91c2d7cb95d80b5 717f16904fcd3dadf21b96c4
FBundlePoint.firstChernCol.eq_1	thm	022421509d0b93dfc3c5d9a5e5cea0f6a9bb47cf 9bdc344be8b7472be029e18f
FBundlePoint.isAdicComplete_of_pow_eq_bot	thm	ca2ce41b1928f1a18ad3407f02242233e91b51a8 1db0cd1038358d4de4185024
FBundlePoint.isNilpotent_of_mem_of_pow_e q_bot	thm	9dadce3f9eef2eedb0a3a0af3af41e975e5adf0c 480acb7f6b5ecce0d5696bf2
FBundlePoint.isUnit_add_of_mem_of_pow_eq_bot	thm	ebb24a53f48501bb79d055294fd7b3b036ad3fab 2ad1d7f29f674c6b83efaac1
FBundlePoint.isUnit_two_dual	thm	1e3c7441099d5d016be06ac03d3a9d43f9e51262 7ce664f7914962c097400f4e
FBundlePoint.lastCol_mul_apply	thm	c454a7adff72cd217d13f31789f21c64c8294303 1e2b220b11bdc9327ef4b5eb
FBundlePoint.mul_pow_sub_of_mul_eq_zero	thm	df98ae75410210484f3df936e60c864aeb8d17f f0c6606f040c8bfe191b60b2
FBundlePoint.mul_pow_sub_of_mul_eq_zero. match_1_1	def	3c06ee6afb10101d7a1fd893ba47f2f6a7b64485 6bddb0fd1f4d0a03c2754e86
FBundlePoint.pow_succ_add_of_sq_zero	thm	e2302c3815819e9944e18a432ea45372c0e783c6 13733e2cb79653a875458109
FBundlePoint.pow_succ_add_of_sq_zero.mat ch_1_1	def	3c06ee6afb10101d7a1fd893ba47f2f6a7b64485 6bddb0fd1f4d0a03c2754e86
FBundlePoint.span_eps_sq	thm	838dbae6f3fc586581455e027caeb88492dbe0d9 49757fe445f2505e8c7ab6e5
FBundlePoint.sub_mul_sum_eigenCol	thm	f2789b125ab9209f9d937b0b6f704ea686a77978 db486ec71663326bc07cfc07
FBundlePoint.sum_eigenCol_self	thm	3750aa0a7cc06a067bfffca027570fe4bf0e18ae b196ec2ae71b867d55d481f2

Crypto.Homomorphic

declaration	kind	SHA-256 of the statement
ZKP.Paillier.enc	def	22753e8eebf79b0514870006645c1f634dc8d457e1b496cd85cbfbf7cb11ef24
ZKP.Paillier.enc.eq_1	thm	4b900bdc82ef075f54816b0ee6ee52ca9c912e6e8aa0806c569c5118ffc41ea3
ZKP.Paillier.enc_mul	thm	6fe777a19b7596cdf4e45062698cba7b16bf0df3db76859f30d05f636ad938e6
ZKP.Paillier.enc_one	thm	42e27c07ce2a3330126e689c6f594a1ae9a929fc4ec5fcb87be5748a8058776f
ZKP.Paillier.enc_prod	thm	f1d45cb47be877d6d739a0c7908b22d706dab1ea1d7533864c0739df1e52b823
ZKP.Paillier.msg_unique	thm	ebba08b1a79fa19647848a870dc9713632627339364a58ba0befc3175d6957a9
ZKP.Paillier.nsq_eq_zero	thm	b483529204a9093f9bb30b7a9b1a7c666535c6d1819b7b0df49e9d02ab7d3f43
ZKP.Paillier.one_add_n_pow	thm	25b2aa8dd53e657ef151f0d56dad7e5bf1e449709c08ceb1e52c602feaab571
<i>Crypto.Ledger</i>		
ZKP.Ledger.Valid	def	a453f8bdfbf8664bec947fd9818ff4aba4d83e5300c7896351224002fa6fae8c
ZKP.Ledger.View.casesOn	def	0c556fe373d9061cde601badb58be52e9436fd9f64d013371f8ec0e0dcde9a6d
ZKP.Ledger.View.challenge	def	347bd0147a5b2b49888bef6e490cf6814b38be9dff5f009f58a90543f5167487
ZKP.Ledger.View.commitment	def	99ec958241b9165874b1e0bf8045a730de11512b298313a175870c0629f732c4
ZKP.Ledger.View.ctorIdx	def	65b3b6610f0c85667a9652abc8dbe6efc01268b46b9ec03324605e68db1e3bb8
ZKP.Ledger.View.first	def	99ec958241b9165874b1e0bf8045a730de11512b298313a175870c0629f732c4
ZKP.Ledger.View.mk.inj	thm	e2a58eaeefba6ee728fb13d7736f8cdfc95a63195028fe21f2f793514fc1a9e0
ZKP.Ledger.View.mk.injEq	thm	0502ee4fe3c91ed2efe0611cfcab758196a3567965acf3a14d7746bcfb7cb71
ZKP.Ledger.View.mk.noConfusion	def	bcd5c5716f1e4539d50c43bb53bc9459b1aab0d85814361d1dd9dc38e539986f
ZKP.Ledger.View.mk.sizeOf_spec	thm	4b0e3b28efeb9810e6b2b12543334361c2b23ffd184c26108f1c7aa452131cf
ZKP.Ledger.View.noConfusion	def	8f07e8f776a71f1e6b483fcecfe3bd63a5901bf0bd887f670159057520b093d0
ZKP.Ledger.View.noConfusionType	def	451c0a71b333507549296421509fa5d0f43cd0eb4c12638c54588a19fc04447d
ZKP.Ledger.View.recOn	def	0c556fe373d9061cde601badb58be52e9436fd9f64d013371f8ec0e0dcde9a6d
ZKP.Ledger.View.respM	def	347bd0147a5b2b49888bef6e490cf6814b38be9dff5f009f58a90543f5167487
ZKP.Ledger.View.respR	def	347bd0147a5b2b49888bef6e490cf6814b38be9dff5f009f58a90543f5167487
ZKP.Ledger.View.statement	def	ff3cc294450608b692e56ff18cd95b4257b7b00462a1a1caaa92acea02eea991
ZKP.Ledger.collision_of_ledger_records	thm	0529f63de028f3f06f1c9f652df040bdddc1fd1992624d44bd911c541128735b
ZKP.Ledger.commit_div_eq_one	thm	40022de351193bca8436411ecfc5328b675ee8476405db8268d3a5f2828a0075
ZKP.Ledger.commit_pow	thm	c1069dd79931c35d38870e8198f47cc30849b9ef e226ad83e433b14196c2d80d
ZKP.Ledger.exists_unique_view_match	thm	419d4e8acfaee874aefc3d43c433622e10d4df47e52583671b29eda6057039fb

declaration	kind	SHA-256 of the statement
ZKP.Ledger.extract_opening_of_two_views	thm	f447f078cef9a8da763bfb3a8b8d2e6e6250287d597648a1605c46f9ddb01f2a
ZKP.Ledger.honestLedger_eq_of_commit_eq	thm	9addadfea13b7446820a484c712b56fb78666004a69943012fe2df0d0fff12f0
ZKP.Ledger.honestView	def	114e873b33d800b5247a025b88998f65b8764c1e43fcf0802822fa9398cbb0fa
ZKP.Ledger.honestView.eq_1	thm	354890090a89063363bc5ec80bc33f07a29cc3e4da6aff7bba01c19a80fd9ea1
ZKP.Ledger.honestView_eq_of_commit_eq	thm	4a8460f69bdf5a009b85cf06e5d9c0d1fd373302b4cdd2f5ce455360b04ef5f
ZKP.Ledger.honestView_valid	thm	e20ec6c39cb27c6912f964c5b1918c7b68ab22cb bdf4d5f6c97a11be6a840ff5
ZKP.Ledger.shift	def	41a7fda205131351bdc572f6a25b948eb6dd702d ee6e0c418fb28e4cb4e76191
ZKP.Ledger.shift.congr_simp	thm	4d4e1c4833a96a8ebd59e4225bb32227cb6ca6d52f5305776ef81edd62003d02
ZKP.Ledger.shift.eq_1	thm	0155df605f61d6415ce7a3750d20755c84c1b3b04144073c1aa685db0b299450
ZKP.Ledger.shift_bijective	thm	2c32726388d55bcec25e724b63c5627b0df107402c8572da0b38def21a9de1c8
<i>Crypto.Merkle</i>		
ZKP.Merkle.MTree.below	def	e3f6ee2c0d6577913b36b278dde17350bf98aade3bc0d3a42eac695c05570f90
ZKP.Merkle.MTree.brecOn	def	8c3278d894d105aaf5c8498a1dbaa52f2733ae2364c7eb5c96654275838b0150
ZKP.Merkle.MTree.brecOn.eq	thm	39dacaf86fabcecbcd511abf0ab3df12d3a89ef6e92137747fb544a58054267a
ZKP.Merkle.MTree.brecOn.go	def	8ad9c10e667a5624f8f8523e4961c02c145c194cd6e4cd28b397c1fb1305cccd
ZKP.Merkle.MTree.casesOn	def	3ce3630d8cc6ff78235f3237a1d33114f9c78d857f1dad6e8364396eb3857d
ZKP.Merkle.MTree.ctorElim	def	cfe35e02c8c9fecc5494c0cea3c282cab7a6dd137b592fcd12778ba0fcc392b4
ZKP.Merkle.MTree.ctorElimType	def	9731be1b4af2660ecb5a85bebe6a1c3f1cf522840016ef5de1597380b0405cc9
ZKP.Merkle.MTree.ctorIdx	def	cf43034e267864631bbe5c58d7f08897e34194bb10fc492dd325c247b53ca057
ZKP.Merkle.MTree.leaf.elim	def	ba8dd13bff02eb8ac3b213d291171f6ed363f1cd878f77925c25d676f4c34147
ZKP.Merkle.MTree.leaf.inj	thm	8a406cea483f328066d7ccafefb8ac9c8c93fa3afb3e9c21080111058d4e87bd
ZKP.Merkle.MTree.leaf.injEq	thm	5c401bf73a2e8267b96a3e0e1e1bbc566cd6eb673e9a8969cd4e4df16c43e45f
ZKP.Merkle.MTree.leaf.noConfusion	def	0df732f74fb7faec8c58c22d6b634f8da97cc9d09fec713fdc82caf74eec8c8d
ZKP.Merkle.MTree.leaf.sizeOf_spec	thm	8c4c119f8bc44acc68ac94ae219d0cc45c07df0c62af5e5ccf75296df0478c68
ZKP.Merkle.MTree.noConfusion	def	be27991dcdf62aab9fe96963ae277dad19e836040c80991c2ff0bfb77be028f
ZKP.Merkle.MTree.noConfusionType	def	9a6b2aa100243fc172bb803c08a90aed9ec80b97c6d3abffa8fbfec93067b9d2
ZKP.Merkle.MTree.node.elim	def	5c815c2cff196021bfcc977dde5342d2913f3e880de9695332b5adb7ffca48ca
ZKP.Merkle.MTree.node.inj	thm	3259885dc7a15112ccd24ba598ad066bb02d33ee4264e45f0782e128aefee286
ZKP.Merkle.MTree.node.injEq	thm	329ef4eef2a7d9e96a240b73fc9d939450a41ff449ea62f54987645124000468

declaration	kind	SHA-256 of the statement
ZKP.Merkle.MTree.node.noConfusion	def	9067bc1b295e214dc867a5450f028b36c56f6c6f116d9c89c660034c22ad3abb
ZKP.Merkle.MTree.node.sizeOf_spec	thm	934ef64b73594efe40116b9444ebbeff6442ea7879e53da66b04d665c78c69ad
ZKP.Merkle.MTree.recOn	def	d430a4c04efc534d7666e30cc13c00c8f922397f725c6ce2bb68aa15e231f7ac
ZKP.Merkle.MTree.root	def	242872e13c02cb03dde816e96ec46d17f4fb51bd218e827de00a5d934a63e3aa
ZKP.Merkle.MTree.root.eq_1	thm	f87ee92c18e47e225b718fd06d921b4646e39cd46feacd83da799ba0133484c4
ZKP.Merkle.MTree.root.eq_2	thm	f521e81d901db8d0c147c6ced5922953ef549b9d79341ae4a179ebd296d95c78
ZKP.Merkle.MTree.root.eq_def	thm	514c98dfbb0adbcab2f42f3a84ed414678742b9119c21895f2772db84aa4e127
ZKP.Merkle.MTree.root.match_1	def	e09d168ff18696e65dabdf1ca295d45495ce8e6e625b79a7fcda0494e4f65798
ZKP.Merkle.collision_of_foldPath_eq	thm	8df4cd541795731c860e343897eb84120aa83dee73a8d1b137000beafe5434f2
ZKP.Merkle.dirs	def	b1b025ebecef21f3a21a9f8fda34c1ec1bb5af543b92fc5cf458f1a6231a24f4
ZKP.Merkle.dirs.eq_1	thm	62552566cc5ffff408abcc3a50b3515384a679ff32986efe07457f61f9bfead4
ZKP.Merkle.foldPath	def	2f536b3abfba3f015c34dbda2dcf23ec75076a5fed6c2181670132bed9c49ff4
ZKP.Merkle.foldPath.eq_1	thm	8891dcb759762e1e8b214f5fab2b7577e3e0051ee65af8be0b81f8661e574dd2
ZKP.Merkle.foldPath.eq_2	thm	22c8467b8a6726172e8ff00a0c9289754ba57da40820f92a2168b9724d32d04
ZKP.Merkle.foldPath.eq_def	thm	126ba0482892ddbdbb5b71e76f350d041f537defda5e1361fe62ac109cf3e339
ZKP.Merkle.foldPath.match_1	def	f4a70dd55d09bbbc5ca80a99d443956d8326dde245a3880b9f5a4b7584bb09f17
ZKP.Merkle.foldPath_append	thm	7f593bb53fcc89383993b3b5d2aa0dfe9e4c7ccb5013a7afc8cfc63f6937cbf
ZKP.Merkle.foldPath_eq_root	thm	c82df551dc859d2e0e0eae9953d0a9088f4953bafd09f52daa979888b4ad401d
ZKP.Merkle.paths	def	2fe90d58106dccbb32d5dd440d025a1c7edc64077fefdf4f7ad77e530cf5e014b
ZKP.Merkle.paths.eq_1	thm	bfe7719d7c2785914e7d00301b3d19f38d75fa5ceff54c315e4bd207b8c34161
ZKP.Merkle.paths.eq_2	thm	d46863ab4ce3c36f9f97f443002465eef768d5aa6622b66e13bbdea5545cc05d
ZKP.Merkle.paths.eq_def	thm	9b58345a1ad2fdee8a4f2a949263f41eb3ded1377a5b213cf2e124722d07ad95
ZKP.Merkle.step	def	788ec73c67bc11c1d0af33b920e081553ec7c725414682ec15c3bbbe8d052ebb
ZKP.Merkle.step.eq_1	thm	c8801e0af8968ec85dee58e752a98e03068a2c9d41046c433a5dd9797a9304d1
<i>Crypto.SecretSharing</i>		
ZKP.deal	def	b92911c954249cbea722673d0eafdaaa0917ec12cdfc388f00f0cbb94cc4e036
ZKP.deal.eq_1	thm	a89ebbb078aa5d8d37dc11a3b9af2a32e7c16728b09de63c01b67e842768db4d
ZKP.dealAdd	def	594c964c1dc746ce2e35927d008559e552763e5ee7d1906889a84601d3686a43
ZKP.dealAdd.eq_1	thm	a9cbac9a076639e7fbdc07ae5a30b0229f1c778b47560c127b0e0babaf7655b4

declaration	kind	SHA-256 of the statement
ZKP.existsUnique_missing_share	thm	e93732a5445805c8514ccf72f98965876039639fd4999143320a7ebd3c606cf8
ZKP.existsUnique_missing_share_add	thm	62d9f6f0033034e77834e3e1185d69c72a132a39c909d08046618d72b47bc754
ZKP.length_deal	thm	c9e5e815f6a44e2152329e4c236e204343651ad7d500769f2137cb4433deec76
ZKP.missing_share_bijective	thm	41a597c2b86afa3172bec2b816b2f6ba1771c3819fea8d3c46d5a858d4928479
ZKP.numBundles	def	3d1917aa425fd7059394e1413a9cf36a1304374e10ea4b88200fd2d34ce749d4
ZKP.sharing_24	thm	9e3e03f988099ced6efe89609da751c51698d979724015787a58c029794ad3aa
ZKP.sum_dealAdd	thm	75165eac80305aefdc959b9735fbecb3791a4d71972aa6eb1474f13e0a3bf5f0
ZKP.xorAll	def	df4612cf68393578a4017ada0fa636eafdbab5b5b67a66a52442ecdffc4beee8
ZKP.xorAll.eq_1	thm	4986dec355587a594f0969c7c5a11635ff48997dec4f867eb78c87a3dde8df
ZKP.xorAll_append	thm	c015b4754ef5a8d9f63f714dbf99ff5f91fde8debfbac460fabec0e83a4fe20f
ZKP.xorAll_cons	thm	bf7b5721b7632137d1c36bb194282be2f81cab0dcaa97403b61ea9b084bd1ff5
ZKP.xorAll_deal	thm	f46082c4f747fdf660ef46ec81a971971db7e1871e3dfa5ae73ba116edb66d2b
ZKP.xorAll_middle	thm	acf24ab9a4f8750320dac652d87941c73b746e2c6916ea898c4db5081493b10c
ZKP.xorAll_nil	thm	b2c02956eaf8862e4eba25d1fa994565a921da37ab2ef3ed9a45b96e927ad91b
ZKP.xorAll_singleton	thm	31da677d32375f1b8ceae797be12808067bb66798c6580dacc3c03e513e87b8a
ZKP.xor_xor_cancel	thm	1e9e693d370507b75934b30c27b681b707fddf94d1a305e4c061f9d66d596582
<i>Crypto.Sigma</i>		
ZKP.Sigma.Accepts	def	22fa852be40eb55dd6d67586633a109766eef1593563cc122d72918832018667
ZKP.Sigma.Accepts.eq_1	thm	bba6d8f577190c99444bf29704a472e508d43136fa9b2066d8783a80cc1fc6d4
ZKP.Sigma.accepts_first_message_unique	thm	838bb894a70c4f3cb644c3eb5bd244153e08bb7d830152527c2940d18f8814f2
ZKP.Sigma.cancel_common	thm	75bf60486f0a2e6a08e7cac01cd2075262718da96fcd054b54c24488fa1b403c
ZKP.Sigma.cancel_right	thm	7d9f1abd0c03d3373f4aeefb8692da728e8058a5ae54c9aa53db0a4dabd12724
ZKP.Sigma.commit	def	fe25330d79a88aa25bccbf1729dfef1d4f59122cbb6a42c49354897b73cf2977
ZKP.Sigma.commit.eq_1	thm	b8ea203462934224386c37765a9c5d387ee23341d8ef8a1019535c688546d5f3
ZKP.Sigma.commit_add	thm	4b8f7980619769ef8036cee3295e01143a84d68097c171643e5f9d2bd4e1e57c
ZKP.Sigma.commit_list_prod	thm	65563bde3624ba591c4c521ba5837f2a412f7462a8488af8fbde27dd559a3dae
ZKP.Sigma.gpow	def	97efaad04a5114b1488b017bf83b3d9c7bd73047c6de86ad7e74735ed74c6a92
ZKP.Sigma.gpow.eq_1	thm	4dd61464a06a3027cafcd8cffa612c47b35f57e228baed0686ca4e51a98c3bdf
ZKP.Sigma.gpow_add	thm	ed633ea2e30e2a413428c9cea48667dfb2494718c0fc04c73132f944c61660ea

declaration	kind	SHA-256 of the statement
ZKP.Sigma.gpow_mul	thm	74e5f93347f6a0bbdd791a56070f0e595161cf992e782c7747ec414cece87155
ZKP.Sigma.gpow_mul_base	thm	87774b4dc5aa8a6a2d00da7ac924f2009799a686e1419ceb6339bb729dd6cda9
ZKP.Sigma.gpow_neg	thm	a1dcffbadc25dd89bbba94d389fa3ef65d5a348cabdeba0f86832876e24e2947
ZKP.Sigma.gpow_one	thm	9aeb696ea1b68a7213aad9025ee9e648f645d1c9e5a83850579ae51b9ec7203b
ZKP.Sigma.gpow_sub	thm	9d7fb0ccc9670d70c485aae7544d154980230d2f0da6fb0387a754dffdd77a52
ZKP.Sigma.gpow_zero	thm	97eae35b766310238937f217ba5ec71ac79727079da12f703413f6a3bfd95ee
ZKP.Sigma.pedersen_binding	thm	951551294093c400a7397fee203fe8dec1f5df07e2524333fd4035a8a864019f
ZKP.Sigma.pedersen_hiding	thm	57be0a396f67325e826bbba2704f0fe736a851a31f8cca85208245016736a81b
ZKP.Sigma.pow_mod_exponent	thm	23cf1186251062b9f3001fb9168f06ae4be80b19ab439e068db31120d48ce54e
ZKP.Sigma.quot_mul	thm	28d64ae96f4ee57becccd41b1fab10e93c7454e969a6c892a3bbe15b97aca8
ZKP.Sigma.sigma_complete	thm	c98e9789b5a0367c12fd62f401c91dce78c6edc145e8f1d82f0a88320ed4de7f
ZKP.Sigma.sigma_hvzk_bijective	thm	ca6b2f027602b13b6890b19b6529e4515c5cbcad3b52d6f9ddc093380ad42139
ZKP.Sigma.sigma_hvzk_simulate	thm	bbaac9d7466d65f8cfd8eff2c998f4c88f2df04b2882a026355191696a3576c1
ZKP.Sigma.sigma_special_soundness	thm	5998a93de8de3a768d9857ec47fd7104628b706ad8b35ec8305de5c9a61ff94e
<i>Crypto.Stego</i>		
ZKP.Stego.chunks	def	acd31563902fad7dad01bce0eda60d60a644a89e1485fe4fa6113fdefe82f463
ZKP.Stego.chunks.eq_1	thm	e09f61eb6bab42397f0bcf7e59e321605ffbd015b80c983c212e9250594b94b5
ZKP.Stego.chunks.eq_2	thm	46d53e8327ad31f6eae174a48cd82083ca29f761bc6b92f3a1782b08b6be879
ZKP.Stego.chunks.eq_3	thm	2f1028340cbc1a79f6ffb3f6c7048274ea0e3eec42aa17d198270ff5c7edcd31
ZKP.Stego.chunks.eq_def	thm	eddb9d29976ff7b495dbf4e36c9d503d43346fcd23146d6bed5d115a73d1a086
ZKP.Stego.chunks.match_1	def	f6037f790f04d2be20f90fe5622d8e8a2b5c694b8062ca5bf24817164cd258c3
ZKP.Stego.dist_setLSB_le_one	thm	fea73a6a9ba4fbf84ff7f2b0751dc8ebf9673cc681197fd3787bd2fdebd0f024
ZKP.Stego.embed	def	a976bc123b858f0fc2c33ba50a162c6f38629cd2cb3adc45718cedf1326f8abc
ZKP.Stego.embed.eq_1	thm	e30f986e7ea7c4969865c266a89b41e0299509cd1cb2f2d0fe8df16fbbc779e0
ZKP.Stego.embed.eq_2	thm	5e9c0e5e3afa71359328408ab8390fcb522583c0e87319477e287e2ab1e047b6
ZKP.Stego.embed.eq_3	thm	e8b391177996f255e4d2a072694e9ec5c356dbfe2c83a66cb771ff19bf416943
ZKP.Stego.embed.eq_def	thm	2e6f752ef6015d076e5ec6251489da83695e334a170c47a9ca380609ae79e6bd
ZKP.Stego.embed.match_1	def	1681728292b183da6bfce7f99b35036f714d3f21f9d52ba0ccaa432862254c7a
ZKP.Stego.extract	def	a5f55d493173d4fe4550a36dd717060eb389ae71fbac735707bfe25ded820858

declaration	kind	SHA-256 of the statement
ZKP.Stego.extract.eq_1	thm	16d72dc141d37efa05d114301ea0d5e2c699ad2b851586e2af36b1b386f1ba0f
ZKP.Stego.extract.eq_2	thm	52f8bea934fa8d34eb76a803358f1d03eb29a22fe804ddac1186dfd22288c05f
ZKP.Stego.extract.eq_3	thm	43690652ffe753435758667741caeb19ba6902c96e155d0c823c7885f75ce08a
ZKP.Stego.extract.eq_def	thm	a1ec95b19e1e8d36b21ca8ad112839416ae38fd9b44d2c7ea55d3504b13977b5
ZKP.Stego.extract.match_1	def	7a1d41326cc1445e1e9cd204af9424fdd6a535b71726c6f6a00234473bd05b14
ZKP.Stego.extract_embed	thm	53e4d0e9079c8e1de48dfd6313023869d67c6961baea4d73a346f2484cf2397e
ZKP.Stego.getLSB	def	c8ba556743a705352d11591b9b0f8bc45d6626725a8003e43a6e911dcb2d78c6
ZKP.Stego.getLSB.eq_1	thm	3ba22891ce859ef40f2ed9c7425c8e8474d3884a6349528725c9dcf17755566a
ZKP.Stego.getLSB_setLSB	thm	5e28808cf97161d334c623bddbecccc0f7c9014c1f9e29011ef3b86bb646f777d
ZKP.Stego.join_chunks	thm	6e748756704c94852ad92107f2fb4540be35df297861f1ffcf34cd6c09ae347
ZKP.Stego.join_chunks.match_1_1	def	6bd230e677f000d2f4a89560d6e01e47afc0b0867c1100d58a433b9b8ab1c1b7
ZKP.Stego.length_embed	thm	8bd6319f65f47cc13a5e1c3010199c16824bdb32d225ccd6341e2dd891989d9c
ZKP.Stego.setLSB	def	a4a4179f857b8ca1d4113fece29be13272dd07ffb4d63beda7d4142db24afcdb
ZKP.Stego.setLSB.eq_1	thm	0bcbe498810bf67b129e2712e785e0608cff40ee5d4ed55f7499e12ead9e947c
ZKP.Stego.setLSB_lt_256	thm	a38d836d9a7451e39709e7ebee56958a92c57efa4b4fbc03e3333878d6f0ef82
<i>DirectionRing</i>		
FBundlePoint.adMatR	def	bceba5489487100a9916b852be5e37413b4f587c410473f0075461ff76c0fb3b
FBundlePoint.adMatR.eq_1	thm	3013e04e6220e73213a8334929b540d180ce9a69a2d8e146720e17a379d0f3a9
FBundlePoint.adMatR_apply	thm	79d1a11ee5cb8fdc4d9fbcd87bafc675dccc19c7ec597ae689b34118209412a1
FBundlePoint.adMatR_neg	thm	dbb5a6f58b2a34282b7e83dd1f53dd2f559c5db12ecee6793987dd045f44ecf8
FBundlePoint.adMatR_zero	thm	4dacec35de4ffe29c926b74fb6d5caed53c8e4a6335ad98e2b02d1262a292ac9
FBundlePoint.derivR	def	f4a8d58c8e6d7d579c906c4ad9ce2e059a8a4fda bc8448689dfb40ec497bc3d1
FBundlePoint.derivR.eq_1	thm	c745e13b19298ebd30c2f3252fef96f51aea8603dc7840c05289e6e616db5f94
FBundlePoint.derivR_add	thm	b3a26d8cbc4f03eb7c63732f12e606fb226420e63e8479d6880a67415a72e0c3
FBundlePoint.derivR_apply	thm	ff7f810e38741def5d86e07d44e391976ccc917c83d6cd2b3cfaf6b189dbab1c
FBundlePoint.derivR_mul	thm	e4deeb101bcbf6e708772cba926153ecf2e630018b4f819089108783fd39efac
FBundlePoint.derivR_one	thm	09de5d9ef366abf6662e8f869f3d9dfffb22630f6e7b7090c204172044e20875b
FBundlePoint.derivR_sub	thm	fb459fa085ebdb2f10f0f9b28c7efe5204a8618bf8ee3263a0d1fa090f5ffba
FBundlePoint.derivR_zero	thm	337e93fabddbcabb0be9d47032f8c0dbc0db1e15ee16997330ce7a4499c2a5f6

declaration	kind	SHA-256 of the statement
FBundlePoint.eq_zero_of_derivR_bracket	thm	4f782e91b141f850170b7929a480438a88ba28e2d83cdafdae357c1a4d5547d7
FBundlePoint.eq_zero_of_derivR_bracket.m atch_1_1	def	d1de05e19bf6fdeaa6a8ace370a0bb1070a429ac b48d8cca9512c4cbd2052ab8
FBundlePoint.eq_zero_of_eulerR_bracket	thm	40ae67c9b29bc45f9410308ecac98c03f7a78acc d2439067d2d7a3a57ea128e6
FBundlePoint.eq_zero_of_eulerR_bracket.m atch_1_1	def	d1de05e19bf6fdeaa6a8ace370a0bb1070a429ac b48d8cca9512c4cbd2052ab8
FBundlePoint.eq_zero_of_isUnit_smul	thm	5b460ffdb59c3ba6e3d893a17ff0a705c8c801e 35eacc55524acd3527209f41
FBundlePoint.eulerR	def	f4a8d58c8e6d7d579c906c4ad9ce2e059a8a4fda bc8448689dfb40ec497bc3d1
FBundlePoint.eulerR.eq_1	thm	4f29ea5b43fdd3dc14daa438d026edef7d7d0dbf 51f0369d5953613a47bcf416
FBundlePoint.eulerR_add	thm	5a49f2141d1731824055eab90a91ec72be308c56 2f4bf31229c182c4a463d8e3
FBundlePoint.eulerR_eq_X_smul_derivR	thm	28f7e68e1c9f0c91e645c6fea5981e6c036d112b 10b2967f0f0d3609f8373ac8
FBundlePoint.eulerR_mul	thm	f4d50928da08a7c861931b973b6c90a02622d795 6ce425a1c1785a38ef23cc3c
FBundlePoint.eulerR_one	thm	3e1fa3f377ded2420cb7c796d71bdfe798e1dc5f 5dd800393b16eef51cc82b34
FBundlePoint.eulerR_sub	thm	b37d2a0d992c29ce6900deabd3c4e5cb831ea4e6 e2505c829d47b4db47848b13
FBundlePoint.eulerR_zero	thm	32fd012ec02d1d38be3d8ad65192a01b8877afee f82c51ddc5746ed3d4e0103f
FBundlePoint.isUnit_natCast_succ	thm	aa61632f0c4763b102fb7955f99d8396a6b7a2cb cfe230f5d1b696978f2649f1
FBundlePoint.psCoeffR	def	a38a8201f9ac7fe6f1c37b6df41187dac5ea09fc 13a0c5f8960ca3d3f00af5a9
FBundlePoint.psCoeffR.eq_1	thm	35bf7e3545092af9075673c5ff172c457f229b07 effc4fe932d4ac5dbd7601e7
FBundlePoint.psCoeffR_add	thm	f6acd7698cdfc49a33879f3757ea711ed360d1be 4adece704df78ef6bab22672
FBundlePoint.psCoeffR_apply	thm	141303a7655884ede78ae87cd06aa7f69900d6df f5a17f63ed5aa4ac00581608
FBundlePoint.psCoeffR_derivR	thm	6866e45d7f7da08c0360d99258e9b033df8204c5 3275804a1bb6e095b1a5e7e3
FBundlePoint.psCoeffR_eulerR	thm	94acde994b4eeb8ae61de289501d0fc7750006fb 70a98dd71f2a5f297fa27678
FBundlePoint.psCoeffR_ext	thm	1e5562dad7b97e52d0f4136a5daea35b6426c9fb 674b054126b6974da8bb158f
FBundlePoint.psCoeffR_mul	thm	59952056253cdbe4ae45a4c430d437dc1d426476 dde8505031a62b6bea4ad3b4
FBundlePoint.psCoeffR_mul_eq_zero_of_le	thm	cf97c6192d3228bfe76d226f74b5d2ac931faa2c 03d4bacdcf0b46bff656ecc7
FBundlePoint.psCoeffR_mul_eq_zero_of_le'	thm	7db1831bf16bca28bdafe8c227a3c1e445cee1d2 227ef98970d3699b7a68f46a
FBundlePoint.psCoeffR_mul_of_lower	thm	edcef61f76d1ffbf55ced75adcc9df175598d8c5 1640848692b01fbc2f7e6c26
FBundlePoint.psCoeffR_mul_of_lower'	thm	22378c9e05c24dbe6e01889dc566755ea51d2e7b 2ebeb2db3105ad08a723e140
FBundlePoint.psCoeffR_neg	thm	14b8dde169ff878cf446781954a7f2f75c28a77d ee8a521bf0d66d8729e9acd6
FBundlePoint.psCoeffR_sub	thm	fbdb69668c9f1ff1403db4f30019a854ca822ed22 5dd4aa5f6f705f78d0de5f05

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.psCoeffR_succ_one</code>	thm	e0703a77eb5d95cf0db501f1f9b9cecc2f7813c0d58ecb0b6d80c46b9f81ac68
<code>FBundlePoint.psCoeffR_zero_matrix</code>	thm	8968d130a5c290e2cab6d639cd833038062af5f182daed22990b005bb0dc86e0
<code>FBundlePoint.psCoeffR_zero_one</code>	thm	52025074c44044d0c53d15191b75eb20aa77117497dc933691233c63ac996449
<code>FBundlePoint.smul_pow_eq_adMatR_pow_of_rec</code>	thm	04e8572e7dae48ae0ce583ea3e3b0897d265968d3d3ce8685c44626f085f9264
<code>FBundlePoint.unique_of_derivR_bracket</code>	thm	9f054fcbebf2df04c96b94e68843f7480d14b9604fa6ab5bbb2389e5f44de15a
<code>FBundlePoint.unique_of_eulerR_bracket</code>	thm	9a3204ba932c7971df19d69e7a89973989ed00f818b8fcb816af9ee22d61a739
<i>FBundlePoint</i>		
<code>FBundlePoint.Fmap</code>	def	9ba6fb243e5891c48158a1b95d7908839d4781c6c625efca97d89f8541b0f040
<code>FBundlePoint.Fmap.eq_1</code>	thm	792618f1f9641912c5f06e5bb112f3c82dee50375313e64473eff3ea082a67f0
<code>FBundlePoint.Fmap_add</code>	thm	037a9745e55a456db9ff6951fdf7f6fde58a436c6d14c42b249c85e3add2df43
<code>FBundlePoint.Fmap_diagonal_diag</code>	thm	425461a3867f383e734f6156b111c18b7ac778f931a5b1d69d4854dda6e9a40a
<code>FBundlePoint.Fmap_one</code>	thm	0a4cea56d5b7039ee1bc28f9f1e3446e48f4355eb26137a88279a66443ef1d44
<code>FBundlePoint.Fmap_sub</code>	thm	1f25417732482b6ceee589cefb7ac803dcaf8bd5baba8d2bd6e80ab5f9bc6deb
<code>FBundlePoint.GaugeSystem</code>	def	17797ce1123ca8c85f350cac6f2182aa5a105589e2bb087a8b526fbb70af6419
<code>FBundlePoint.NormalizedRec</code>	def	c820e21dfd254945954ef3017fda04b638e01671829fde22e8aa6840920f7257
<code>FBundlePoint.adDiag</code>	def	9cbc346c080d87efb61ee3a1125d891a31e461094bfd01f2f0fed472add01dfe
<code>FBundlePoint.adDiag.eq_1</code>	thm	a9df9043b2d50c17616733d835dbb94f2134aafffa1d51a6e8891e6534329b88
<code>FBundlePoint.adDiagInv</code>	def	9cbc346c080d87efb61ee3a1125d891a31e461094bfd01f2f0fed472add01dfe
<code>FBundlePoint.adDiagInv.eq_1</code>	thm	1e1e07d9c7171780c160e77ac52b4366705f31cc682adef878ba0d204763824a
<code>FBundlePoint.adDiagInv_diag</code>	thm	857593290595c0c8e2c1ceefdf4b51842d5926c02ff45d14728253df052db827
<code>FBundlePoint.adDiag_adDiagInv</code>	thm	3ab3061a2f4bd32fe191752c975501c360ae3bdb53f7069fd35c8f326ba2bb63
<code>FBundlePoint.adDiag_add</code>	thm	dd132b2edce0e6c62ed9d200899db739fda5c10087b5906784ecb597513d9270
<code>FBundlePoint.adDiag_apply</code>	thm	4148df33b83e423a512867d1f08417aaa2fa5f447469aa2b5c7833e3c411215c
<code>FBundlePoint.adDiag_diag</code>	thm	043d3c49b6b392e066ef8a76c259372c7060a78a34118514c80d1345e9ab2f5f
<code>FBundlePoint.adDiag_diagonal</code>	thm	6b189b38e9ee2288d11d46042af9470cbdaf8b0a86dee7c00b5aa0364e10ddd7
<code>FBundlePoint.adDiag_sub</code>	thm	b9c1dd5b65d898c69398905809eaf1403a6b0e1d8ee4b66d4f0ae699ac7f6d24
<code>FBundlePoint.eq_diagonal_of_offDiag_eq_zero</code>	thm	93caddbb62c08425eda21ccb60f3da279dc2f9ae90225a6cab6aa5c66a32dd3a
<code>FBundlePoint.eq_zero_of_adDiag_eq_zero_of_diag</code>	thm	3b42db338aba8f9bdd37fbb2b3d457559e57b5992d418dc5aac931976de36311
<code>FBundlePoint.existsUnique_gauge</code>	thm	cdcd5acc4c12c9e74c30fd8ba2b748a1567cd1b3d74668776a612c7a7501d7a5

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.existsUnique_normalizedRec</code>	thm	bfa3efc741c9d598e9a66d819ef0765dde165cd5ef2dee15fb9a361d6e8ab2a1
<code>FBundlePoint.gaugeSystem_iff_exists_Q</code>	thm	0b8c37729f7e963d226b60eb4d0eca2ef0ccf448404d7e58aade853e8f8053b9
<code>FBundlePoint.gaugeSystem_iff_normalizedRec</code>	thm	29d1ec09fa6ad4b353146d3e1501e08b8a2f4b220330bf0f3359bdf4a48898dc
<code>FBundlePoint.gauge_conditions_of_gaugeSystem</code>	thm	ace2885744d998edfdff5cf79ec76fff8e9a9401c288007c4a9349c987799f45
<code>FBundlePoint.isDiagonal_of_adDiag_eq_zero</code>	thm	20bc5a192da37acce9d53d4fe42fc60496720b73e38687fa2614c108d29284d0
<code>FBundlePoint.mul_right_cancel_of_isUnit_det</code>	thm	ccd39a0d37b5f4dc63e55b09b0cb5619d6388478c895e44dd9158d7fbc49bcff
<code>FBundlePoint.normSol</code>	def	36a6e87deb2d8a1e2db8053f1428bf4887b3012267623edca76df02933288baf
<code>FBundlePoint.normSol.match_1</code>	def	e24e03d60f53a6c058b653b20b41aff674a44c44a810b816e23592b63f683138
<code>FBundlePoint.normSol_rhs_diag</code>	thm	16d1c823ba8afbd2f73343839b782276df0cccf140474180df61fc5a46e74360
<code>FBundlePoint.normSol_spec</code>	thm	4c03f97f79e618a0c957cc020c01bf24a3355bb74c17fbd24610b496d70e9daa
<code>FBundlePoint.normSol_zero</code>	thm	40ab4b43b38f1d0eaaed05caa01bb395ccbe4c6714381d91a27c745b599de57
<i>FBundlePointPowerSeries</i>		
<code>FBundlePoint.IsGaugeEquiv</code>	def	57ebeeedcd91ec635000a00e23e0f06978323dc2b896c89f32cc419719ea2f1b
<code>FBundlePoint.cst</code>	def	77117c4a8a61e4ce4c558db56fa5a39f8f36a17f3776e79222ad23337676c199
<code>FBundlePoint.cst.eq_1</code>	thm	6f56125e26273f92ed5f2f4f70e8233eebb22647f9e082121036e9cd96b990fb
<code>FBundlePoint.isGaugeEquiv_iff_exists_Q</code>	thm	edcd50bece77b15d88e9594487f6728714e8ae2a893f95fc678591a5bfa9ee17
<code>FBundlePoint.isGaugeEquiv_iff_gaugeSystem</code>	thm	5bbdcf918761d0265b06083e87e87c2bc831a150a3d16fa794a93dc8397a9fb9
<code>FBundlePoint.isUnit_det_iff</code>	thm	fd7afe9dbfbccf702e04e72c8d8f519318af2a065d84aa9f986c4dd557af169a
<code>FBundlePoint.ofCoeffs</code>	def	babd8c3839c611817a92c9941551242514b42ac17809377ea40f85d35f4f3fff
<code>FBundlePoint.ofCoeffs.eq_1</code>	thm	af24915a4540b80648e35b8db99523d189e4194d689d5cf55ea316fa8e7f7db2
<code>FBundlePoint.ofCoeffs_psCoeff</code>	thm	8abf44701e5b1321fba4d2c0b797912204e69c8a66c34686f763939782fe29c9
<code>FBundlePoint.psCoeff</code>	def	a740baba8f807818312c9288ae03bf752e8b6162ff249e2a115a81e9212843cd
<code>FBundlePoint.psCoeff.eq_1</code>	thm	629727836b0ee143eb7a1315b3e48cc7e8d634d34d1b58e921618b9c727a53b6
<code>FBundlePoint.psCoeff_add</code>	thm	bb57ea8add05fe95c6062a84a890df3f5f8f14d2587e2f27fa8be3bc7e20046c
<code>FBundlePoint.psCoeff_apply</code>	thm	31455921a01cbb0ad4dc91c8f27a22a6a9c516a6d9d53a93bff9dfaae38e4ed7
<code>FBundlePoint.psCoeff_cst_mul</code>	thm	d361fd0e5aadf374884f70974cf3df42af895bede4c465a8071e11b376bc5d64
<code>FBundlePoint.psCoeff_ext</code>	thm	108f0d159c26acd3b85d956e8c8106069a641fe12cc3c14b839d04c564f7a066
<code>FBundlePoint.psCoeff_mul_cst</code>	thm	455975fb45b82c8222edc71fc0ac72bd4f38cd2402b00ce94db86e04b32d9ae5
<code>FBundlePoint.psCoeff_ofCoeffs</code>	thm	07426bdd609356f9349be045b4655b9170ea8342c29fe87687b244ad472d57f6

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.psCoeff_succ_X_smul</code>	thm	084a8e23c5de919587744dd6dd76ab31496b6cdb080dc3166cabfe8df16c8813
<code>FBundlePoint.psCoeff_uEuler</code>	thm	d018cd17ccc2182b8a09588ec166d9ea55829df1a7ad18f0729057242000e6bb
<code>FBundlePoint.psCoeff_zero_X_smul</code>	thm	fbfff130ebe1195ef4ccdefbaa275aa4f4451addbbd718bb7b9fc2082ed1aed0
<code>FBundlePoint.uEuler</code>	def	d850dd658d7f608ecc6607ad8c08d69ae1f830ce689bf57ee0199927a3aecfd3
<code>FBundlePoint.uEuler.eq_1</code>	thm	efdbf8d2d219d066f6d16a677e92845f61762c7750935db1be66dd1e44a7c213
<i>FlatnessGauge</i>		
<code>FBundlePoint.cst_zero</code>	thm	41436d013ea5272f6a94628792e23fa1b2ad0247c10f8d29ed1f8b8cc710321c
<code>FBundlePoint.curvature_gauge_uDerivT</code>	thm	23e91d92456cef2b7befd2d6ba3f643e3c25dd420574028bb691774168794f82
<code>FBundlePoint.curvature_gauge_uDerivT_eq_zero</code>	thm	253dcd4fdf7f1c4f967e5ed70dd1038fe3922f443b7759dbbbb33de2b51da8e1
<code>FBundlePoint.curvature_gauge_uEulerQ</code>	thm	c63d3de17f2391e31cbb133384bcba707f577bd9e5e03ad5dd5dbd925bf3d223
<code>FBundlePoint.curvature_gauge_uEulerQ_eq_zero</code>	thm	6447bc6d3fdf663902cb89b8d390253261711ea3b6db61cf59e6afb6c3121b4e
<code>FBundlePoint.derivTu_mul</code>	thm	5d037563f438fbd00ea85c9f18a6f070ae0b98a8855b9cde7ea8677585fd61bb
<code>FBundlePoint.derivTu_one</code>	thm	3af10282da6514d602ec30ee7b20ce2d107e3ce01a68490c34a7cc555615fb32
<code>FBundlePoint.derivTu_zero</code>	thm	dca218cdba541fb63f5e819ce353a5718b53814436c02ecdea9ce37b6564227f
<code>FBundlePoint.eq_cst_gauge_of_curvature_eq_zero</code>	thm	70ebce949ff8f423e701f1fd18722a5300e21445c6ff699537383c77016bab54
<code>FBundlePoint.eq_cst_gauge_of_curvature_eq_zero_q</code>	thm	f504593f5d0f21133c04b6e8a20635649b6c881a1422bdcc7f7a8e7cf9247886
<code>FBundlePoint.eulerQu_mul</code>	thm	1aaecb21b1e4e63484f619dd003ea7d4eefa1b6d72d9be7425524b48619ce54
<code>FBundlePoint.isFlatAgainstFramedQ_iff_curvature_eq_zero</code>	thm	57f796112935cefc60ff9721602a69c14b7ac92db60e8cbc27322d125f689199
<code>FBundlePoint.isFlatAgainstFramedT_iff_curvature_eq_zero</code>	thm	4a36fe673de496e026fbfbfbebaf051485fa9b4f8bc1e1e8abb4db101e0b6b0
<code>FBundlePoint.psCoeff_succ_one</code>	thm	c864c6231eb7b5853e4aab36d20538ff3a0c135c58aa6eb1882451817c891202
<code>FBundlePoint.uDerivT</code>	def	4ffb29666621e5f317b75a7080e5bbba364c52f145d25ddfaa6558e122aab653
<code>FBundlePoint.uDerivT_apply</code>	thm	ea789b5487f13dbba20e896567df0804b446026176b3f78f28221cd071138493
<code>FBundlePoint.uDerivT_leibniz</code>	thm	ff4b2955b0a3188b5da1a05fb217da01b68079ec4e2006111c7b01d6af1a1217
<code>FBundlePoint.uEulerQ</code>	def	4ffb29666621e5f317b75a7080e5bbba364c52f145d25ddfaa6558e122aab653
<code>FBundlePoint.uEulerQ_apply</code>	thm	4ade0eb39762a45181621193d8b305267082797c56af6bf1e7a5e0239b3ef915
<code>FBundlePoint.uEulerQ_leibniz</code>	thm	525d00e8adf59f6c6c0111547670f8d6276d69f9d948408afcb7f8b36a136d36
<i>FlatnessPropagation</i>		
<code>FBundlePoint.IsFlatAgainstFramedQ</code>	def	39f0d13308a4da71a7df9f5df7f9565f5af6f0a916f451eb1df8d29ea59f2346
<code>FBundlePoint.IsFlatAgainstFramedT</code>	def	39f0d13308a4da71a7df9f5df7f9565f5af6f0a916f451eb1df8d29ea59f2346

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.derivR_psCoeff_of_isFlatAga instFramedT</code>	thm	6681ebc66c51b3cc9fece1a14aa1626284a8c4ad d04959f595f16e41cc479bad
<code>FBundlePoint.derivTu</code>	def	63b22e1e9b85ccc9c6721aa4785df23ef7ce1063 8842e2c0db8a0146368c52b1
<code>FBundlePoint.derivTu.eq_1</code>	thm	fffe56f407748e9bcdeb335121c809f2749afdb0 ea3509781702e3b4c8df80d6
<code>FBundlePoint.eq_cst_of_isFlatAgainstFramedQ</code>	thm	4c39441a6fffb6e0ed5c02442d0d84be6e20f1b7b 21cf7e128471126d59fe1328
<code>FBundlePoint.eq_cst_of_isFlatAgainstFramedT</code>	thm	e6668cce5ada1a2613287d2b1692345cd7c359fd 7dcbec222b1d7c17f541bd3f
<code>FBundlePoint.eq_cst_of_isFramedDir</code>	thm	0aabcb60148ebae2cfb92ad5ffc4db33be0361a5 91d6f14b7d0ba944050f4369
<code>FBundlePoint.eulerQu</code>	def	63b22e1e9b85ccc9c6721aa4785df23ef7ce1063 8842e2c0db8a0146368c52b1
<code>FBundlePoint.eulerQu.eq_1</code>	thm	55750186b3ce575cbdf14ffa4cbfe8c9230992b8 2ca367ca17e8a280dceffd2b
<code>FBundlePoint.eulerR_psCoeff_of_isFlatAga instFramedQ</code>	thm	6e51ee741f148e8e32a194295c851e62ec011694 860a63c6c446f82937d5902d
<code>FBundlePoint.isFramedDir_of_isFlatAgains tFramedQ</code>	thm	5021ed6872c37a7f05ab12449ee85c87570634d5 e14c3391e98805a9010b57ff
<code>FBundlePoint.isFramedDir_of_isFlatAgains tFramedT</code>	thm	f53855b09353e4770a857fdd21c5fcf099e86a5a 6ddfea7423b86b1c1a11a3a5
<code>FBundlePoint.psCoeff_derivTu</code>	thm	99c643be342b579f68df7d7fe11ee5422e141a94 706369153a34fa89825b928f
<code>FBundlePoint.psCoeff_eulerQu</code>	thm	77a6ae31016ba522cd0b1ae4a391c6d96873047e ae1bbe8f5d941d0e29850f74
<i>FramingPropagation</i>		
<code>FBundlePoint.IsFramedDir</code>	def	75875f7d566658921498760973c6083166a148c8 22be29873ddc23c308a6bc04
<code>FBundlePoint.IsFramedDirAt</code>	def	75875f7d566658921498760973c6083166a148c8 22be29873ddc23c308a6bc04
<code>FBundlePoint.isFramedDir_of_flat_q</code>	thm	3e0d700fed2bfabff80059387bf28d2aa1e092e6 052be09c061bedad0c82d86d
<code>FBundlePoint.isFramedDir_of_flat_t</code>	thm	54c0ddaa72ae7ae59c92aeeb0dede016f3aae604 26c42ef6f562fb91fadd5184
<code>FBundlePoint.isFramedDir_u_of_framed_q_and_t</code>	thm	4108dec8b85731b6ecc4d0341cc947b6bdf140a 74a80d68ace73cd092e144a3
<i>GaugeCurvature</i>		
<code>FBundlePoint.curvature</code>	def	1684dff1d9cfb6a10eea115aec4927750fe02048 d3b61f6ea14f2dbae9b58bd1
<code>FBundlePoint.curvature.eq_1</code>	thm	9dbcce14d3ed267c729169c06b6ea4e073824473 e96de96b2b8ac6b945735352
<code>FBundlePoint.curvature_gauge</code>	thm	b6084a09f9ce870c3c948a89e268ad2b817b14f0 fa58432f758eba198bedccac
<code>FBundlePoint.curvature_gauge_eq_zero</code>	thm	b514f69f4ccf29ab1d69db792629272148b9fdf1 e40e7abb3b72c406b5185f40
<code>FBundlePoint.deriv_inv_eq</code>	thm	5beea728d78109874c65d7b5fc9a3c31a523eb1b 76b38d94432f15c769cb7091
<code>FBundlePoint.gauge</code>	def	cc2a907b50f6531b1dd544721636b8bbb1d4ba3c fa2dcc803cd59092c063fb7b
<code>FBundlePoint.gauge.eq_1</code>	thm	bf87f3a20d36a79586b0fb9415af37e6cbf5c5ca 6a2ce8d1f86865354b38e15a
<code>FBundlePoint.map_one_of_leibniz</code>	thm	a684d497398156dde4e41cb51c114bf006f1e561 b47d5f83d3be4eb9b65ff310
<i>MatrixMultiDirection</i>		

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.adMat</code>	def	3a0c0070c6e8261dd5c67207bb7bcef7ebc046b4f1c0f126f42674185878fdeb
<code>FBundlePoint.adMat_apply</code>	thm	bdf23323fa1fd19755b98354849d41f839360d5dabc0eddb5b1e21d5d2b398b1
<code>FBundlePoint.degree_pos_of_ne_zero</code>	thm	56295a43e5779bb1cae441187fe7ff8049b07706a33952c0a8d9b2b7c761bc4d
<code>FBundlePoint.mvEuler</code>	def	4f3184b863a9faa5c4dc355aeb9b763b977649faa8a2a5abcbf82285f73e251
<code>FBundlePoint.mvEuler.eq_1</code>	thm	9bf7657362e8643c1f335a6551d39069b267d32e78ce94037c6ead5a59d63050
<code>FBundlePoint.mvEuler_smul</code>	thm	44094141d65928dd54e511b1b8f11178d42545c6e4b01c64bc61d0069639b232
<code>FBundlePoint.mvEuler_sub</code>	thm	4a9a1d00d10259ecafa15a18f9c902165ef3469b425e759bcae0ee68a7d62d79
<code>FBundlePoint.mvPderiv</code>	def	4f3184b863a9faa5c4dc355aeb9b763b977649faa8a2a5abcbf82285f73e251
<code>FBundlePoint.mvPderiv.eq_1</code>	thm	879dbb9775e87c9c57487f267f821eeadece64f0984ed9f03117343942a777ae
<code>FBundlePoint.mvPderiv_sub</code>	thm	9bd3c2d62a86319280021bf3186174d07a649899e367a06cc3e29083c8f8b24b
<code>FBundlePoint.mvPsCoeff</code>	def	3a7b11f9b63d1efe3cd84ff93b900aec8fc855c3da7372daf78bb01f989d2b4b
<code>FBundlePoint.mvPsCoeff.eq_1</code>	thm	da940791d5c7bfc212ba4a799f7769275d3102550f891d6c958d2004e15c68d3
<code>FBundlePoint.mvPsCoeff_apply</code>	thm	5a149235a17c61168e74aa0cbb088e052ebb95a0a6696fdd0f7a189d282f11c9
<code>FBundlePoint.mvPsCoeff_ext</code>	thm	4ef0c6ac9df304240509cbe7d29d79a271b45b72f6af99a04178eaa3490754b6
<code>FBundlePoint.mvPsCoeff_mul</code>	thm	5387b38a67b1c9e6ecb02f41c08cde0719eb9246ad1b90dada3ce67a79fc687f
<code>FBundlePoint.mvPsCoeff_mul_eq_zero_of_le</code>	thm	df80961105659f0244358ea2a5c67b7a2cf9cd5004231fa0e20725ddd10c59ec
<code>FBundlePoint.mvPsCoeff_mul_eq_zero_of_le'</code>	thm	88348fcd974490e110c27ec488a0a8a36751a532c6afaaa42110c3b5dfef599
<code>FBundlePoint.mvPsCoeff_mul_of_lower</code>	thm	c11802c59531d6f3cb46ccb424ae9f041674d3a0435efec18f481c4a4ee66c53
<code>FBundlePoint.mvPsCoeff_mul_of_lower'</code>	thm	98a04cbe9454fd45a066b636bdf014bd26aa0d0c50dbbcb1a84394a13141daa0
<code>FBundlePoint.mvPsCoeff_mvEuler</code>	thm	92b60eea2f2f72d62fadaed4629de3a127533d39ef58858e6a8a7c535dec52f4
<code>FBundlePoint.mvPsCoeff_mvPderiv</code>	thm	b327685d4f262c542d1d354af57fd1a90051b895517bdb0ed94490154c1c98fd
<code>FBundlePoint.mvPsCoeff_neg</code>	thm	4d8def8e0b66aaa354ae34890684564530c78953c41dc5f82677cd5ea898e17a
<code>FBundlePoint.mvPsCoeff_one_of_ne</code>	thm	351577b63ede0b45cf4f218cbea053b80e96345b555e8f9698e92caa14c38c9d
<code>FBundlePoint.mvPsCoeff_smul</code>	thm	981a0462f1aab0b70bc567a079382b8f8a35f542ce1567e35ffae31f4cdb3ddd
<code>FBundlePoint.mvPsCoeff_sub</code>	thm	d0ed789b98676c7bb530dadfc3417f0ef74381b2104f17904c872da2d4247ea0
<code>FBundlePoint.mvPsCoeff_zero_matrix</code>	thm	1c7f910994fd15f76d4528027b5deeff9332847220452d09b8b99b926effa2f
<code>FBundlePoint.mv_eq_zero_of_mvEuler_bracket</code>	thm	ccc6a47bddfada124d148b9e747b255df50c6b27d64523ec2b5500fa29bf9d29
<code>FBundlePoint.mv_eq_zero_of_mvPderiv_bracket</code>	thm	95015ad81f23320f2e630ffd342555541d700332365e795bae8df4c87feabef6

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.mv_unique_of_mvEuler_bracket</code>	thm	8d75d12ca8845606effd5c00219e0f6dc502c36814a90d7c3ad08034ead46b85
<code>FBundlePoint.mv_unique_of_mvPderiv_bracket</code>	thm	5f72ab50d60c1e2a000f464eb8b28628f25ec762eca6df893339a3b9ef034056
<code>FBundlePoint.smul_pow_eq_adMat_pow_of_rec</code>	thm	1e8f5eae5e25bf843832008ae83b09f1f3e9387121cb4de39e803a60efc0676
<i>MatrixTDirection</i>		
<code>FBundlePoint.derivPS</code>	def	d850dd658d7f608ecc6607ad8c08d69ae1f830ce689bf57ee0199927a3aecfd3
<code>FBundlePoint.derivPS.eq_1</code>	thm	1e4765f88cbcf8b19ffe8ba5be2b708145e36d04731cbec98c6899d1616e955
<code>FBundlePoint.eq_zero_of_derivPS_bracket</code>	thm	5ad0be2b5bf4ca182f9540b87587a8725561d27e65c9a6b02b99239430e8e79e
<code>FBundlePoint.eq_zero_of_derivPS_bracket.match_1_1</code>	def	af482b7b311e61c5702d559b58e6b87ad41021bfe60f110467924c19d93bce40
<code>FBundlePoint.psCoeff_derivPS</code>	thm	5b79434907f061bbc4657fc5236a11686fc7165389e03df9bc0e06e61bd7574e
<code>FBundlePoint.psCoeff_mul</code>	thm	fc4bf8765b613d27351bd77417e602731110e78d324e2701b48d62d74fd4bd62
<code>FBundlePoint.psCoeff_mul_eq_zero_of_le</code>	thm	bbc8bbb54ac4274d4c6fd1a4374a3ea08a19eda21df5eeff4abc7f2b76593214
<code>FBundlePoint.psCoeff_mul_eq_zero_of_le'</code>	thm	a9860df63e6404007d75ec644c141f0bb226958fe998395181ceb2e86adc0165
<code>FBundlePoint.psCoeff_neg</code>	thm	a55fae4ee0d689cd0b785d2bfff89be845fd2460757f1338b645f6f26c5a6d6c
<code>FBundlePoint.psCoeff_sub</code>	thm	aeb77a727d3ef5defafb18a4229615db281ec0fd099f18dc6725eb624c7e5b13
<code>FBundlePoint.psCoeff_zero_matrix</code>	thm	37a891d84443b02a263d44314e976a68a2511f658acc0e6d23cdc0b5c9b4f18e
<code>FBundlePoint.unique_of_derivPS_bracket</code>	thm	fc1dcda0a5b1ba32cf05992a0339ef9ef3ab1ceec19a767e9a68f6e4babb8e36
<i>MvIntegrability</i>		
<code>FBundlePoint.commute_bracket_of_mvPderiv_bracket</code>	thm	44bbcc3e240447aaff127c9a23c45492903f801e988828756dd46949bbc81bf0
<code>FBundlePoint.cstMv</code>	def	8068042d4f7cd50cd7917d38fb0f2fa870edc0dbf1b10b0d7a9ce9ed8956de35
<code>FBundlePoint.cstMv.eq_1</code>	thm	ef37eb0e1a4edab0fa8468aa9c5ee53b02f1d4f69d783bcd92f9f27e2452aaff
<code>FBundlePoint.mvPsCoeff_cstMv_mul</code>	thm	9e6a456d287e06487b99f2da3977ddf660821d6e45bfb898793e9f05766f2f22
<code>FBundlePoint.mvPsCoeff_mul_cstMv</code>	thm	2469da44149fdc4644f3cfbdd8433c34a5d857508e1f2fbc36018fd9ab742067
<code>FBundlePoint.mvPsCoeff_single_of_mvPderiv_bracket</code>	thm	d9644af6efbccef2295e2507597f808e2d8f89d143defaffc8188f81fea00ffe
<code>FBundlePoint.not_exists_mvPderiv_bracket_example</code>	thm	ae568936aa5da421993138b68c5f28d7d48273fc8784d2875d204ed87264141e
<i>Nilpotency</i>		
<code>FBundlePoint.adMat.eq_1</code>	thm	5b89f3bc8fb8a28afd9aa4211e238e2d15b36a29b5a2ce7251e4838faad60885
<code>FBundlePoint.bracket_pow_eq_nsmul</code>	thm	da66ed335dc3ce2c406eda35ff2a6776e194bd15bf570534c4b5db4206b6ab91
<code>FBundlePoint.isNilpotent_adMat_of_framing</code>	thm	9e6135588f2af1df8190ae3b307d3c44ddcc0386eebf8a3a5b949994705989f2
<code>FBundlePoint.isNilpotent_adMat_of_isNilpotent</code>	thm	8ea5ccd9e34e506a71e4f93c0480bcf83fcffdb2bd90d85c873e357d0eb398d7

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.isNilpotent_of_bracket_eq_self</code>	thm	f8231d30c2c2fb4413e21a5e5ed0ea29414a09290e065228ea1ac79631faea41
<code>FBundlePoint.mulLeft_pow_matrix</code>	thm	615d95c5907d676145069729413d3616bc9b4f12cf1a20709fa300fcca96a0dc
<code>FBundlePoint.mulRight_pow_matrix</code>	thm	986be2ae280d0f011666f3b79b369cda51858b7f98894258e768e160c783c938
<code>FBundlePoint.mvEuler_eq_zero_of_framing</code>	thm	59a329775ff2780badb3c0bb9b449be21f78fa10e86754d756ab38d5bc14c1e4
<i>ProjectiveBundle</i>		
<code>FBundlePoint.cexB</code>	def	dd78cc755d40b750cdc927084f172d4ab1daeef557ba7bb11253606c64bf65b1
<code>FBundlePoint.cexB.eq_1</code>	thm	7a85c20551cb00b31fe2ec413308b651442c10c08fd4c1407f9ff175f3d11159
<code>FBundlePoint.cexB_isNilpotent</code>	thm	11029bd81bc0478afb3a5f0736c8398922064533eaba30da9c084db2be654835
<code>FBundlePoint.cexLam</code>	def	d316a8badd44c3f5e9e5ec64d86e48bbeaf24054749cf78421f219070cc20bf7
<code>FBundlePoint.cexLam.eq_1</code>	thm	ec348b46b0f362d2939bddaa64df46ac1679b6dc55ac57638f287342215c0c29
<code>FBundlePoint.cexLam_sub_isUnit</code>	thm	b1cbf148244e701361dd63919f00d152e950a9957da32ecd12be58f1da861710
<code>FBundlePoint.cexMu</code>	def	d6491a7aa93b1b75520aec5a72bc858a6059e6dabd09dfab8033842532cb8d5d
<code>FBundlePoint.cexMu.eq_1</code>	thm	b0e7034f4fb9e0baf23c9cf79a5b69026fe8038d525de34ae21d27e492473a06
<code>FBundlePoint.cexMu_sum</code>	thm	a067195a5940bd18f8deda58d8fd2d2cd0f5aab2c269a0a7b10e2d3367bd29cf
<code>FBundlePoint.cexP</code>	def	210af4b786c3c8f3744ff6f0e3a04a6c52a76b80a11731977f1d956ed20174c7
<code>FBundlePoint.cexP.eq_1</code>	thm	801f50836990b6282af24d03275eca451a98bec7555b7dde108158b31ea65c63
<code>FBundlePoint.cexP_mul_cexQ</code>	thm	492106c17096cb03f5eff0f9e63415cee7cf49ea35547a7b595fccb37420bad6
<code>FBundlePoint.cexQ</code>	def	210af4b786c3c8f3744ff6f0e3a04a6c52a76b80a11731977f1d956ed20174c7
<code>FBundlePoint.cexQ.eq_1</code>	thm	9d558f4882b08e66a1f1308689a1ac97dde5dfdf7c7090c4140bfa4ee84eb9e
<code>FBundlePoint.cexQ_mul_cexP</code>	thm	b6fc4e071083e3e0937cdedc0c37bb5b7ce3d8d5cc37eeb92a346f5bc245a948
<code>FBundlePoint.cex_diag_entry</code>	thm	6e540ba15f04dde2d45d746f9be245448c8fab84ea9716e0e161dce65e43b1ae
<code>FBundlePoint.companionK</code>	def	72b48aef7b589718b35f32352ff5ced3a279a751358cf015ae3e04ef1e7ba816
<code>FBundlePoint.companionK.eq_1</code>	thm	a5dab9cffa9f93ae27a024bf836f1ef756c8a5ff9d0f6296c002c1b8ae4148c5
<code>FBundlePoint.companionK_cexB</code>	thm	134671dd9fdd632e9df1aa1afe7ebb3ef00d443def70d9c932f70b0e2cfce5a2
<code>FBundlePoint.companionK_mul_cexP</code>	thm	34ca7f740a6074f0047b9cc8317a718d4b9800d616ad1556e594d533b4c26f1d
<code>FBundlePoint.companionK_zero</code>	thm	0e476fdb59ecb8787663b0905820f44a8d69bf72382b515bd972125f30960f04
<code>FBundlePoint.counterexample_diag_conj_diagonal</code>	thm	d861c1191c3e73efb6c2fe6db44061434da0dc2b951762155c37100420ed43a7
<code>FBundlePoint.counterexample_diag_conj_diagonal_of_any</code>	thm	64786fac10d3459913378436ca9f0424b311307008577cce2b1ec08d06f7e702
<code>FBundlePoint.cyclicShift</code>	def	1aa8182b6a9714c5c03a53442960d94e751a3c0d6d62be1bc922d52d23f2c234

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.cyclicShift.eq_1</code>	thm	8cacd969dfbe5940ca3fa9eb97c903ec6da16ce45ac113fcea7eace7f5f5289b
<code>FBundlePoint.cyclicShift_apply</code>	thm	f8b75164d6bc20067916b1cc8588884d24647dfa62bbf65771b03a33b1367cd3
<code>FBundlePoint.cyclicShift_mul_apply</code>	thm	cd66c43db6e8d36210269dc4893edc7dd03db36d4c6eccc17de9323a44f601aa
<code>FBundlePoint.diag_conj_diagonal_eq_zero_of_cyclicShift</code>	thm	feba15703a2a58476b97e78d45f1a1509accebc2cee3692e83d259a86e3c32f1
<code>FBundlePoint.diag_conj_diagonal_mem_of_companionK</code>	thm	5e503ef060ceff1722eae4555b35c6a873e4fff46d94773c80bf2cef3affb72b
<code>FBundlePoint.diag_conj_eq_of_diagonalizes</code>	thm	399a017ebb0b53d9d5a88d916aa47b8e2711e74b4de83d8fea83dfbe226019a8
<code>FBundlePoint.diagonal_mul_inv_of_diagonalizes</code>	thm	3f4314b95afbd042c405ceb6c2bf2a470a00bc241f05df87d3607ec6a054e17
<code>FBundlePoint.dualNumber_counterexample_data</code>	thm	f50d59e99ca1da914e118cf725deba60c3724aad5567fb8733a1389206be2e89
<code>FBundlePoint.dualNumber_diag_conj_diagonal_ne_zero</code>	thm	cba2fca9feec724039355f764553dfe4baacfa6d1dfeb081c20b8db829c66509
<code>FBundlePoint.dualNumber_diag_conj_diagonal_ne_zero_of_any</code>	thm	ae1f1bd4ac9b319b613a0eb162da42e4d8ec289da021a8102d0b3e3375900858
<code>FBundlePoint.eps_mul_eps_dual</code>	thm	506403b79245d4de38b48320ca0fcdc06164d39035978f9e169f3a82a061987e
<code>FBundlePoint.eps_ne_zero_dual</code>	thm	cf2c1589993da12031d7d3ae7f847dada1acbaafc156f5b58b2d7a2996a67bec
<code>FBundlePoint.eq_diagonal_of_commute_diagonal</code>	thm	d74efcd1c39c2b07a46ef918d58daf5c7a63c6cc2d882e2a011b23865fcad1f6
<code>FBundlePoint.lastCol</code>	def	72b48aef7b589718b35f32352ff5ced3a279a751358cf015ae3e04ef1e7ba816
<code>FBundlePoint.lastCol.eq_1</code>	thm	2066fdd74d4beb019bfaa740a53ff9efc25a352ba24819d81bd3cbaabf08200e
<code>FBundlePoint.mul_cyclicShift_apply</code>	thm	cc2eb7b2291a728d550e1f3aa76b283c902a88114eb30a378443cdb01687efb5
<code>FBundlePoint.mul_ne_zero_of_add_self_eq_one</code>	thm	20e45ba6863ca07dd1da94fc6f6b9a82a1f3e55887974218080bac115e322e9a
<code>FBundlePoint.qHalf</code>	def	1974ba873f5550ddd2bdc2b45452f9a987214fdc6e199ccccfb812a1020142a
<code>FBundlePoint.qHalf.eq_1</code>	thm	8af30e9dcf79d548a764dcad82ee6ff23c4f6cfe577ad9203c199cb4b9cc9630
<code>FBundlePoint.qHalf_add_qHalf</code>	thm	c61971b01127521e660cd8729e407946c33f5e8a2964716ac8e400765c80d322
<i>QFramingExistence</i>		
<code>FBundlePoint.QFramingSystem</code>	def	8d2fe14a199ab4443054ad373394a7d9498bedf70faf40c329c7d39de9975264
<code>FBundlePoint.cst_injective</code>	thm	87a989f38a1c3449a821a78ac2471e8cebc64bd8c2ee1d7049f56ee4bcb2d399
<code>FBundlePoint.eq_iff_of_sub_eq_sub</code>	thm	e9a0a7abdb93c7a8e094d102429922b5a544b762afc041e6d84bbf05fcbfcc7b
<code>FBundlePoint.existsUnique_qFramingSystem</code>	thm	f4809e39054f44eb46762e6232b834d84cc5b0fa29ae994b3af8fcf8845cd256
<code>FBundlePoint.psCoeff_succ_cst</code>	thm	e7693e7fa66648173cabe5bf4210bd2f80695e6f50fff6fba0cab3bc65c58b9a
<code>FBundlePoint.psCoeff_zero_cst</code>	thm	aef5b50ba4b5f7f4b994fe0065380debbd13dfd89d6b0032681c275a5654d830
<code>FBundlePoint.qFramingAux</code>	def	fe857bc214455b43ec1c4fd7d134c8cc2bdf0767b73ea95c63cc83ce4111f886
<code>FBundlePoint.qFramingAux.eq_1</code>	thm	421262ec541b7cd490ff6fe925d8d68bf0d8c498e265b922f67191760e13f1ba

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FBundlePoint.qFramingAux.eq_2	thm	a2c09c3261b07060071535e771f5ea2657e54dae7cc7469a947f6066f2c77d6c
FBundlePoint.qFramingAux.eq_def	thm	edde636b77cb4e4d671d76c2583e942829694c98535a1577b06e965d2e9846c9
FBundlePoint.qFramingAux.match_1	def	52d0eb6c42572858dd0767c6141fae90019d12278bdaa57490569441b3aa333b
FBundlePoint.qFramingEq_iff	thm	97c9272ef474c8666955d3d6d1005a5744dfdf42697306b407594490f509ea99
FBundlePoint.qFramingP	def	8dcb4499350126b24a4f6d4cd670499fff7bc86bb3e6f875eff2e49fc57f8058
FBundlePoint.qFramingP.eq_1	thm	cbfbd9c921f53fbfe0e533534ffd1df02e06fdd2df2aba79d6fb529ceeb438f5e
FBundlePoint.qFramingP_succ	thm	f8d3177fddff2d0f16f220e088bf056c0a0bd43f7849b76d3e7c792706ae593b
FBundlePoint.qFramingP_zero	thm	0b3ed70e66bc2c2b871cddc7dd7a2076975f5c01ad9fa8ba1bb6d1bc3d98694f
FBundlePoint.qFramingQt	def	a1e1474db451a0807e20e968e121ac6805053a3e14e1a7451dd03dd8e1c8df8e
FBundlePoint.qFramingQt.eq_1	thm	6485993b6e8b80ae2b29a4a98cc0737dfb4d08fbfea9c8d2eb6e3c34f94f246e
FBundlePoint.qFramingQt_succ	thm	1022d386b6fdeb7d6561545b69158d29459dda6d0b96b28302600ff2454221c9
FBundlePoint.qFramingQt_zero	thm	f1fde4f2345d6bab582935020b95307edf81dbef4a092432c8a4655cd6c90a07
FBundlePoint.qFramingSystem_qFramingP_qFramingQt	thm	4d7047043c9ab7361979eefb5e12a11b7356b77dcec460331d34a039b3204d5
FBundlePoint.qFramingSystem_zero	thm	5837bc07274dfe07aa7f6e35331c5ffc26f2c7f794b003998cad8a60dacdcfad
FBundlePoint.qStepP	def	fa96327d6623f227aa73d09ec906c03298efa4ad11554aa84041b08ea0a72e9b
FBundlePoint.qStepP.eq_1	thm	f0c8c231e31c55e05770486cdf60ec47e6208d17d49a3731290f4cb13c80cd3d
FBundlePoint.qStepP_congr	thm	6eadd7f3cf383d8463d8ac946a68558b18819d87311547b28109a6610fefd576
FBundlePoint.qStepQt	def	ad2525cdee9f142b5ab875bc49baa9ec9c1f14568fdb22ac1d9f4abc172730a1
FBundlePoint.qStepQt.eq_1	thm	331d2bb7d511d94a6898533ddf96e30601e66420055f1a31b003c1b45d8ae323
FBundlePoint.qStepQt_congr	thm	e2593ad318d34f68926c041173007a003d6b8c5911195b1724bb30e5eabcb9ac
FBundlePoint.qStep_spec	thm	723fbfa3d3d7b03901b30f0dc43eca6fdb6a62588e7c4c62079815655d2aefa1b
FBundlePoint.qV	def	e972f3bcc5f38d8d4a86f683befe509ca23e3a2a2b3a2362420ca7ed521b8a1
FBundlePoint.qV.eq_1	thm	db8531dcf04ce5dbe4edd7e2e9d42b23f1b05b102e91ca3534a26ebd6aa57dbb
FBundlePoint.qV_congr	thm	6b68c756a51565fa20795b0e968faebe33232cabd5835b1545b82858132ef597
FBundlePoint.solveShift	def	991ba9522aa254af24e81af4573eb9d7dcacf72c7dfa51481455efbb3039cedfc
FBundlePoint.solveShift.eq_1	thm	2dcf55d8330480106c371e8daf01ec6039523027a782610ce0875d979bc70c7f
FBundlePoint.solveShift_spec	thm	80c98201401e49fc13c60161f00dc55c26f4ba5c09cfd3ef18534bf6bb0f79e3
FBundlePoint.sum_antidiagonal_snd_split_fst	thm	157199ba7533219afe3cd76f871fd7566a7017924d825b93a88449c49e3f56a8

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.sum_antidiagonal_split_fst</code>	thm	f5aeadf6505f4c5e47674b54e970b4df291be9767d46ee5557a3c79f86d07ff9
<code>FBundlePoint.sum_antidiagonal_split_snd</code>	thm	74e04785127ad576eef28be68db6ea50770b1f233398f404a0eb391f5f48f946
<code>FBundlePoint.unique_qFramingSystem</code>	thm	51fbaf61637f11416488bc52479c4d4d489488f2c4861437a85416982514f03e
<i>QFramingSeries</i>		
<code>FBundlePoint.IsFramedQ</code>	def	b2eab65d00299b6a0268b031a49b3b018e7e6055c9c6abf899f306d75587d8c1
<code>FBundlePoint.eulerQ</code>	def	63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1
<code>FBundlePoint.eulerQ.eq_1</code>	thm	3d45b0d6ac910e206dd7c1692f669c62e19ae5fd9026c58e5519590abf1555a9
<code>FBundlePoint.existsUnique_isFramedQ</code>	thm	8bf40f46edb4e31e1e1e51bce481fa105c30437579f20e20bfaf7807e207408
<code>FBundlePoint.isFramedQ_iff_qFramingSystem</code>	thm	31565c8a26ceee2da2d135ad15375673fc6da8c7ba84e16119bf04799327c333
<code>FBundlePoint.tCoeffM_eulerQ</code>	thm	6c50490ef09cdc4761bf48157eb92b12368accd3284a8d9244ab443d786e34c6
<i>SpectralDecomposition</i>		
<code>FBundlePoint.SpectralGaugeSystem</code>	def	795c00f368b2de9345efb1c49af757adb8c81f0f77747864ca1166d6a6dc1d8a
<code>FBundlePoint.build</code>	def	eb4803dcd79de81bab3aafbde3800dcded2efd837b3e72215810e90faf36895a
<code>FBundlePoint.build.congr_simp</code>	thm	09c98ce7717f385611d4def5d8cc03b6a3c1f71f2ad50681daa8b734d4d62142
<code>FBundlePoint.build.eq_1</code>	thm	4a2eaf5a831cc7574e91696428937f184820fd42426b88fff04b2f4408b2dad7
<code>FBundlePoint.build.eq_2</code>	thm	f2cf5e7dd95d951d086cfef425d0ca70773a25bac68bd8faa367b82be7397311
<code>FBundlePoint.build.eq_def</code>	thm	17e6b4808c1af9b243c7416628b01bd9a185db392e2df73f09d7ee375f9a4cdb
<code>FBundlePoint.build.match_1</code>	def	52d0eb6c42572858dd0767c6141fae90019d12278bdaa57490569441b3aa333b
<code>FBundlePoint.build_fst_eq</code>	thm	888c08ae2183bec7bd6631cb18fe9701d7decd37ade48c246435a5a1a41c1b5a
<code>FBundlePoint.build_snd_eq</code>	thm	fa2fb910d9e1bf8a93a90b099a823e45ce44cb294e0ff8ec1399b35206c9b756
<code>FBundlePoint.commSol</code>	def	6f60091ad80ae42d96021e77a2fd1d243984c924303392bd3c58cf7495ba051a
<code>FBundlePoint.commSol.congr_simp</code>	thm	a66850523ac483ce97b69096c3711216f5fc3f122a047b04b89269c9fcc6acf5
<code>FBundlePoint.commSol_spec</code>	thm	54b10a726e18b1354f9f090d1e192d920b48451a16241191d70b93573e018470
<code>FBundlePoint.exists_spectralGaugeSystem</code>	thm	5be18c1e82d0c5bc2d62d6e9d772c3f29f583fb1eb38ffafe785636294f1dc5b
<code>FBundlePoint.exists_spectralGaugeSystem_of_disjoint_spectrum</code>	thm	3d51cdeee5b180a51cf99b87c0b86d788d024b14eb0da4c02f0a013bc11b4a4b
<code>FBundlePoint.isBlockDiagOn_specG'</code>	thm	d9562e2f4149151184c043b2cf2ba5b4d9af11d84d2bedc0b67349e0d9edfccb
<code>FBundlePoint.isBlockOffDiagOn_commSol</code>	thm	b334fc21ae462f9441143a8f186b8deb03e73da456e0f721527039d7deff8f77
<code>FBundlePoint.isBlockOffDiagOn_specPhi_succ</code>	thm	48696ac88940d302bcbef3f0496f501ec02c4167f5825beb7bfbed17b4685e59f
<code>FBundlePoint.specG'</code>	def	4bc3eba3c91153e9c5392a052a7fe5478772c31a7b86969cb3cc29ebe603e314

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.specG'.congr_simp</code>	thm	2892e8043042e3073b8c989ab844b49bef31c337cb7d6b7d954aacad1664e8c6
<code>FBundlePoint.specG'.eq_1</code>	thm	183f2fc13f6f1adf898d424e01be0738479b7359393a6321974604111402e29d
<code>FBundlePoint.specG'.eq</code>	thm	0a9d0aedc9a8b893c2b0a0bcfb4aa0268030150e9ac6f49dd6da369c57dad321
<code>FBundlePoint.specPhi</code>	def	4bc3eba3c91153e9c5392a052a7fe5478772c31a7b86969cb3cc29ebe603e314
<code>FBundlePoint.specPhi.congr_simp</code>	thm	684582eea946f1a0aba1b93805ecc9e88ca29320945ffbaecb02a68ecd30d7ae
<code>FBundlePoint.specPhi.eq_1</code>	thm	a3e5287da5001d90760f17afd2c3b2ec0d8be2304dbbfe29311c970cb7e381c5
<code>FBundlePoint.specPhi_commutator</code>	thm	baaad9167e292eb66301193546d7c7c22398028e388645b01b112368d6edcede
<code>FBundlePoint.specPhi_succ_eq</code>	thm	15db3085dfe6f4f1a2e35395e472a2b6f76d660bc3c404f293162bac660276cf
<code>FBundlePoint.specPhi_zero</code>	thm	96ec4de7bdec413ac0d68d644b4b422b08c9f8e25bccfbc5b94fe6ce2becd73d
<code>FBundlePoint.specRhs</code>	def	745224ced14eed056d8bc7f71bb5798d941f86c0fc6223d87cedbef30db1022b
<code>FBundlePoint.specRhs_build</code>	thm	33c6d3a5f35d9a03ff33f1772609863978e9e2baf994a8524aa35f29d38382e7
<code>FBundlePoint.specRhs_congr</code>	thm	3315e660734ebe2bad7858ced507e87d25b6c8831ffd0775e94b3bafd611dcfe
<code>FBundlePoint.spectralGaugeSystem_unique</code>	thm	dd218ee2a76d3c7f85d3604ea7941a583dff6fa07ec79609e5180b6f145e8473
<i>SpectralPowerSeries</i>		
<code>FBundlePoint.IsGaugeEquivSeries</code>	def	25baa5837717acd053bad691a195ab4fa86d565d3156147b48bb1659d25ed333
<code>FBundlePoint.cstM</code>	def	7e7065ec3e758793fd7ab83bd44051b5e9599f64ce874075d375387f441e976a
<code>FBundlePoint.cstM.eq_1</code>	thm	5bb06d2e2c21f7dd7a3aa1de7a26dde060ff3e5088f984108613589f551ee648
<code>FBundlePoint.exists_isGaugeEquivSeries_isBlockDiagOn</code>	thm	34c2a280231d8f6b61fc6b51588b3f8f795d62fb089c6395302511274318003
<code>FBundlePoint.isGaugeEquivSeries_iff</code>	thm	34b7a241577d4c4cf4a9d9ef0746d03a1babcccf72d7addee0274b1857e4ec7d
<code>FBundlePoint.isGaugeEquivSeries_unique</code>	thm	22c389691bf534a58ddbef370d5125209189935f6ea4657922b9fc3e885876b8
<code>FBundlePoint.isUnit_det_iff_psCoeffM</code>	thm	6273cf91504ef070857cd5d1bc2e787e89ee67edcb1ee1350a56b5221b750f58
<code>FBundlePoint.ofCoeffsM</code>	def	ac6efc65f1bbd42a3a6700af6b9003032555502211db5d5d1acdb851f3e36c4f
<code>FBundlePoint.ofCoeffsM.eq_1</code>	thm	cf49ba5f378eea8bb402dabe87222cf2b3c7bbe5895fcd9003f6589b8bfb3b17
<code>FBundlePoint.psCoeffM</code>	def	0d338cb7978dda0ea5509c2314bed9b007702fa6bcf9031c3ca062be4a43e3dc
<code>FBundlePoint.psCoeffM.eq_1</code>	thm	7e225569df6b279db4a817f9676d0096b04018df8ed002f726a04a5d34b7d70c
<code>FBundlePoint.psCoeffM_add</code>	thm	e8010acb723d676bdf9c64e20e317dac4d19776c1225f4864a57991fa3aa8d79
<code>FBundlePoint.psCoeffM_apply</code>	thm	c5ce15589a8b4794d13215fb781c063f192743a3625051e3c327769d9926713b
<code>FBundlePoint.psCoeffM_cst_mul</code>	thm	f9c4bd405255316c58d859401a4187f69cf9bf01e6d07c07f7b74e51d5c54ced
<code>FBundlePoint.psCoeffM_ext</code>	thm	5d64f6b3add1841097816a6064008600d6179bc9224ee3c9cdf1dbf85a8cddcc

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<code>FBundlePoint.psCoeffM_mul</code>	thm	f818e3ec262c25f01d6aee8e3614d90f477cee62bf064b77e85fc3750712f9b2
<code>FBundlePoint.psCoeffM_mul_cst</code>	thm	00e4c408835496f4c08a049f8c4776916921ae3e8ea94457febcd1e6792fc944
<code>FBundlePoint.psCoeffM_ofCoeffsM</code>	thm	d81feb1326a99564b0de409a7452360790737ab180abb9c124d5c612be02f735
<code>FBundlePoint.psCoeffM_succ_X_smul</code>	thm	d244b54218fc7d0b60e43417a45cacbc291b393d16504cae040313ebbcc43c00
<code>FBundlePoint.psCoeffM_uEulerM</code>	thm	53a80d613842af08abbef161a328dfbcd704d89cbbdb9631dc79b82b4df4131b
<code>FBundlePoint.psCoeffM_zero_X_smul</code>	thm	701aaf279d5e6b06da7170cf45abc810fdae88207bbb46baa3885eb213126dec
<code>FBundlePoint.sum_range_succ_swap</code>	thm	048158e74dfd047379afed1df35362abdf70e4dde75a2dd645cea989a6faa10
<code>FBundlePoint.uEulerM</code>	def	bbe4c53a46746e439dcc51e765383cb1f82978a3b4dba317d39075fede03cb68
<code>FBundlePoint.uEulerM.eq_1</code>	thm	87be47a55e5308567f3facf7f68e29d7903d0f066cc2c6e660b06f1c50524c9b
<i>SplitDirections</i>		
<code>FBundlePoint.IsBlockDiag</code>	def	814219426fe307dd3c4d4fd2a4b5e146669a9db27ea36b31f1f4efdec8b9843f
<code>FBundlePoint.IsBlockDiag.add</code>	thm	182cf5f8ec701d608a195d1acda57fad918ea6aa0713592eb4f42f0624ea2e7c
<code>FBundlePoint.IsBlockDiag.mul</code>	thm	1f19be51d91e493e46efdd35c7138849a864b4613c61c035a5a632a9c5b6fde3
<code>FBundlePoint.IsBlockDiag.neg</code>	thm	7e198aad926be2ad4477518566d3c98df25b0b47e7820766aa1224d5b1f9227b
<code>FBundlePoint.IsBlockDiag.sub</code>	thm	b4d54455c2206c91c2cc4a7b0194fef059485f0f924927f3f0566a2d60e10a51
<code>FBundlePoint.IsBlockDiag.sum</code>	thm	2bdc27882cfe87a87347464cbd736a483e80553de9a106664d71423d698fe0f6
<code>FBundlePoint.exists_sub_commutator_isBlockDiag</code>	thm	c27fe57da794eca94321cb1e0ad6a38c6341ce6e3195ae3abe361f8aac007e23
<code>FBundlePoint.isBlockDiag_iff</code>	thm	0fd16ec91298b2c2428655d4cdaa8612fb98e2a711bea748b508d24dacb88f88
<code>FBundlePoint.isBlockDiag_of_commutator</code>	thm	ff5a554e795a70da0ec43d2706bde9fdbc9da70ed33a911154cb3f211402709f
<code>FBundlePoint.isBlockDiag_of_flatness</code>	thm	b1909e23d3e56e9592329fdc465a98065837cc5eb6d5183a03d56b5d7654a360
<code>FBundlePoint.isBlockDiag_zero</code>	thm	3f06a6e3d9cb7f5b6160f4f22256e90f435b5254fb1adfd10e5de9b28479677b
<code>FBundlePoint.toBlocks₁₂_sub</code>	thm	7b592eb56055713f17ea0415e91cc504991b46ea2ddc1fdc41aef2f311f2b9c8
<code>FBundlePoint.toBlocks₂₁_sub</code>	thm	1fa316308ca222dd601f5bcd6c177db8a878dbdd868d9bfb138b1fc7d1493eda
<i>Sylvester</i>		
<code>FBundlePoint.adOffDiag</code>	def	5f4d8e3471372928ced11c5277418800760efc69ff7f846d9da5e33b22464f5d
<code>FBundlePoint.adOffDiag_apply</code>	thm	e3ad6a53137e7aef8a5da951f606233b5b04613141ba1008c00fab69f8e925cc
<code>FBundlePoint.adOffDiag_bijective</code>	thm	2ccf71e55566ed77087519aee0d4856f0f07397ba95da687729ad2e6c257634e
<code>FBundlePoint.adOffDiag_bijective_of_disjoint_spectrum</code>	thm	e12a45491f11405fb9bcca5af6fa9165977fa49a3e431d2e263094f9780c389b
<code>FBundlePoint.aeval_mul_of_intertwine</code>	thm	2acbd707ceabeaa7c2424f30c600f3a64d340092d1d864b96a6982925543e7df

declaration	kind	SHA-256 of the statement
<code>FBundlePoint.blockDiag₂</code>	def	cccb9b2b6ba9a926e6fa8c89ced013c939d42b0d2ea34992d1f2e884e1060622
<code>FBundlePoint.eq_blockDiag₂_of_commute</code>	thm	bcd649c5cb66e2488efb4a34e6d16e598c4abc6daacb97f6903d0bc51eefa396
<code>FBundlePoint.existsUnique_sylvester</code>	thm	39616398e33ec309444299e38ab6bf0ddde593bbf761bb55f395c330584ca1e3
<code>FBundlePoint.isCoprime_charpoly_iff_disj oint_spectrum</code>	thm	189efd7521ead4ed96123b7229e0d26e848a7fe537b35bf193b6146400471f52
<code>FBundlePoint.isUnit_aeval_charpoly</code>	thm	65c5ea55eb87b167f06454c5180f2970c91c9596dc2849bf5b62463dd651d728
<code>FBundlePoint.pow_mul_of_intertwine</code>	thm	cf00075a537c537fc92b8262c7c3f6a1e94516d8b09d6a87eb965d62ae1a658e
<code>FBundlePoint.sylvesterMap</code>	def	b211c53acdaefc4deeb9f297d05f9cc9eeabe0ab3a24554899e45f1e88be6a85
<code>FBundlePoint.sylvesterMap.eq_1</code>	thm	2499cb4c1ce9fd274c444da4bd9eb7f16fa01923b07bad497d775ca249cbf1e1
<code>FBundlePoint.sylvesterMap_apply</code>	thm	7f70b5a0740851d76243a095c36a2967a2c571a0036df53f4985d7701c65e480
<code>FBundlePoint.sylvesterMap_bijective</code>	thm	4ac0ee478018e3a21a6c00cd9733af2649f537b333bf7b8a98269d2e6b1ee679
<code>FBundlePoint.sylvesterMap_injective</code>	thm	40f05c464becbd23acdabf4ee0c96521871b5b03f72e9f3b62dda31af07d80a8
<code>FBundlePoint.toBlocks_eq_zero_of_commuta tor_offDiag</code>	thm	27f02e53d586c26d27f5a732790118979072e7de96342cb0fd47c4e81f6bf190
<code>FBundlePoint.toBlocks_eq_zero_of_commute</code>	thm	bf0ba462a00bc03be0eff7591e2c6f09b7588c2d7a297161886b79143d31d15b
<code>FBundlePoint.toBlocks_eq_zero_of_disjoin t_spectrum</code>	thm	7c6b07a609ac117c1215a29848d0102fc187a8ec17c463a8ada70f0cae04aecb
<code>FBundlePoint.toBlocks₁₂_commutator</code>	thm	333050e2f2e52dbaf1905a6631327e8c1db4f7bd7c6881554f170d0353edf297
<code>FBundlePoint.toBlocks₂₁_commutator</code>	thm	46d4eb0ea30b03fc4d04862c8f9c547b3033d533463471d1e63a7e0f01993a80
<i>TDirectionExistence</i>		
<code>FBundlePoint.existsUnique_derivPS_bracket</code>	thm	65cffa963e101fcb95cbfdd0550a17cef2b9b07c0bfbcb8d1bc7bf9a9e19478e9
<code>FBundlePoint.exists_derivPS_bracket</code>	thm	00cbd671a9d8f142118c9c90d62c8dccc13284e0f96ad10fdbbb29cb61c54229
<code>FBundlePoint.psCoeff_tSolution</code>	thm	cc026bc9e4c1e8ed3dbbb3ce1081c681c079384aa92087bccb040549a44f377b
<code>FBundlePoint.psCoeff_zero_tSolution</code>	thm	d311d0212bd941496c48709c8880a5650ab938f8a6c367c5eb9b18b040cf7cb6
<code>FBundlePoint.tCoeff</code>	def	b6d42119081bf99c9642d7716cf6597cdfc41efd07b01814bbbcbce414c0f66f2
<code>FBundlePoint.tCoeff.eq_1</code>	thm	2bb87427b1c18cce0b4f117985583873b592b5b99755f0d2cc9905657bd1ac1f
<code>FBundlePoint.tCoeff.eq_2</code>	thm	051b84034687553e3e16d6e6a66223c93e42f6cc7823d7b1c508696f8f98c629
<code>FBundlePoint.tCoeff.eq_def</code>	thm	3ef5b6f60f54d026f2c771433d934bd8f5644f081871ea509b8fcd22fe9d2c4
<code>FBundlePoint.tCoeff.match_1</code>	def	a53dbddf92ca72478fd3ca3018af758758e87f16cb081960f2a91904554cbeab
<code>FBundlePoint.tCoeff_succ</code>	thm	4f518183994ec362d240fdc866c4ef48585250fbdef563910135f7061e60fea1
<code>FBundlePoint.tSolution</code>	def	71a576c2369ebff95c9bb4b75925ea420d713ffd8c90060a29a67901d8582071

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FBundlePoint.tSolution.eq_1	thm	e54cee6461d4df0de67722b78747c34ef46e3fc89b519873473f33238cebd772
<i>TDirectionMv</i>		
FBundlePoint.eq_zero_of_derivPS_bracket_of_mv	thm	cdd74267cb7877b19cf4eba63e96cef330b7c783b605100201adde304585e76d
FBundlePoint.finsupp_unit_eq_single	thm	e6b9c962f9826f7ef646c2cf24780f982abe22ae0aa18a805a244d83d4d19c12
FBundlePoint.mvPderiv_unit_eq_derivPS	thm	0456d5e10449554d0faa2fe74207b6496c199dfd14963f9379acf663c3e57558
FBundlePoint.psCoeff_eq_mvPsCoeff_single	thm	37055350e41818a608e8cb15ec5be4fc5af19cf3303e51d8db99aba509000fc5
FBundlePoint.toMvUnit	def	f0f388fcc3418260df7fcc1cb138c3e293c0ace9046c965e0745381dc8dc8fe7
FBundlePoint.toMvUnit_apply	thm	6a59b6d00bb54f2475732c03278329aaf8a25ab279fc817c9a6e7d64628fca7a
FBundlePoint.toMvUnit_injective	thm	2ca4bef7e32d40cdad0615d49451ba6c44bc5b43b65b61197b244a37381808a0
FBundlePoint.toMvUnit_zero	thm	642d2b48cb189942f64e03a93eceb5e6644ce3ac99a9e353d0aed2c9eed28bbe
FBundlePoint.unique_of_derivPS_bracket_of_mv	thm	bb7299357d12e99ce9565d69e8d5bf94762b15d6d4d6d8c450aee2a230ab9094
<i>TFramingExistence</i>		
FBundlePoint.FramingSystem	def	8d2fe14a199ab4443054ad373394a7d9498bedf70faf40c329c7d39de9975264
FBundlePoint.X_smul_divU	thm	ecaf94cf88aa846c925495d1d1e1ba293210fd9d35bdbb96c0cbc9409f4bda6d
FBundlePoint.X_smul_injective	thm	4aa4852d5235acdb0dcd54e5c7dbbca05837cfb5988bb5f00380aaf8d2a58cd
FBundlePoint.divU	def	f4a8d58c8e6d7d579c906c4ad9ce2e059a8a4fda bc8448689dfb40ec497bc3d1
FBundlePoint.divU.eq_1	thm	ac86bc813fa55972fff41d79ad2a6f03cfc0cacd45d8cde4ecf4f76771c618b8
FBundlePoint.existsUnique_framingSystem	thm	34096b0fb13b5b82aff115922b0c96d44d6622c95569d8e68ce4820b1f0c7512
FBundlePoint.framingAux	def	c5cc871ed07e7cf725455c328214a681e830e888c7d1cd7674bee0ec589bdbba
FBundlePoint.framingAux.eq_1	thm	0a76f375d57619367914d07b50a588bf1ef0d426f2ac3a050aa24d9ac4e58076
FBundlePoint.framingAux.eq_2	thm	6c26177d324997e43d594dc81a27cc61a5527547eaac77e58b0b56ac2e292460
FBundlePoint.framingAux.eq_def	thm	0e4e3a1cb9132007dc3096b18b4fa8d139ef8d5a01b562e5a5daa7a567bd95db
FBundlePoint.framingAux.match_1	def	52d0eb6c42572858dd0767c6141fae90019d12278bdaa57490569441b3aa333b
FBundlePoint.framingD	def	e972f3bcc5f38d8d4a86f683befe509ca23e3a2a2b3a2362420ca7ed521b8a1
FBundlePoint.framingD.eq_1	thm	f63bcbb0bd6640a6f54e130e1c2011464fb6f76cd34a2f3ad0c02d5986c6bd9b
FBundlePoint.framingD_congr	thm	4b5fc755ac68226be8522cf757cad2b28ed5272fa77cb2b7fc2ac6a68ac8eb56
FBundlePoint.framingP	def	33a642df9e986d92a4331b12bac96cad689b461298ce2eafcc39b05381172295
FBundlePoint.framingP.eq_1	thm	3d50c4fe4772fad0f607bb3dac922b2a241b4a3d491d1877f009cc74aae2667b
FBundlePoint.framingP_succ	thm	7fcfaaf2991e11682e2ea6bb57a7b0a4579c779291266172f4bfca1bbe3394b0

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<code>FBundlePoint.framingP_zero</code>	thm	5dcc1cf50c5dad31158a91d75a64a22ef13b155d6f58a6f55d725b2140a03b96
<code>FBundlePoint.framingSystem_framingP_framingT</code>	thm	28a8fdb2c4e7ea9a327d07e7606f0a70bd30aba01f72a4da09f7bec68c288ca3
<code>FBundlePoint.framingT</code>	def	27117864f6c1c871d136ba801791f1b690fcb283354b1e40f5b8d808073ecf5a
<code>FBundlePoint.framingT.eq_1</code>	thm	c8a43e31b98ac567e117a4487c6ac11c0bee946f59a7109ec05f7462456116e6
<code>FBundlePoint.framingT_succ</code>	thm	4d59249f3521403ecc95e58581565e341c2d9419e30980276dbe795cfda9063d
<code>FBundlePoint.framingT_zero</code>	thm	0162ab05133572d51c699cdd6429129b145fbe88baff2315ff39c6203be1afec
<code>FBundlePoint.framingTtStep</code>	def	edbb5dc9a3f1fbb1865288656e13420430f0c4d0f8cdcb755e5dbd67e73ec29f
<code>FBundlePoint.framingTtStep.eq_1</code>	thm	2ee2772f3c76910119356af7441a9bd381a7b4f04db6d0585cb0443157d3c097
<code>FBundlePoint.framingTtStep_congr</code>	thm	bde31c820dfe060f839607c5e16b146b8407428fafb5c0524f2e42d0b84e1e9
<code>FBundlePoint.invSucc</code>	def	d386a6037bd33dbf61bf75ff4d49b09ea8bfd34755274ab4dc6fb0e84c6a6b5f
<code>FBundlePoint.invSucc.eq_1</code>	thm	935d9f6091648ef89b8f22b063dee8183cc261e1dfc9a0779802726090929757
<code>FBundlePoint.psCoeff_divU</code>	thm	af932b5635a8161ba6f37adcb5ad0f479d4e00bb04808111c4e3c2cb7cd17e12
<code>FBundlePoint.psCoeff_smul</code>	thm	2d4ae8b7cf828b7eb5a5f06bc5325855c3be8849e189c17527c643e7ccd7477f
<code>FBundlePoint.psCoeff_sum</code>	thm	efd344f2c6a64ec738f413aab35899c24b94cccf536ae226022f46073847b9bb
<code>FBundlePoint.psCoeff_zero_framingD</code>	thm	8e3f1db5d8d1c02a6762888df840f3d245866f07fe159a22520a847ab0b98b9
<code>FBundlePoint.psCoeff_zero_one</code>	thm	c76a527cbc87f4e06dec9dc3976a2e51e0368611c9d3516612ced8eca3b86b66
<code>FBundlePoint.psCoeff_zero_sum_mul_cst</code>	thm	adc5c230746bdaef929c791e035a9820bde74e1ea632b9e22b74245fbef314c5
<code>FBundlePoint.smul_succ_injective</code>	thm	2fc79f37dc66b82cf25d4d6bdb2e9c6626f026e19818ef8fd434b82df7e51cd2
<code>FBundlePoint.succ_mul_invSucc</code>	thm	5cb67726420999e9da279464d60f74c39be28b6bf9d847dbc8b3d2e5be4aa2af
<code>FBundlePoint.succ_smul_invSucc_smul</code>	thm	75ce30a713f73d2801f3aa51974587643dd140fed795b71d5d74fbc63bc2c85b
<code>FBundlePoint.unique_framingSystem</code>	thm	8b8ae4b912c5341680aaa9fdff7f5dd5192b72ae1327e0f10c006c35c5f0e5d1
<i>TFramingSeries</i>		
<code>FBundlePoint.IsFramedT</code>	def	b2eab65d00299b6a0268b031a49b3b018e7e6055c9c6abf899f306d75587d8c1
<code>FBundlePoint.cstT</code>	def	45475d8fe1e9a250f2d6ac8e92034c9bebdcd5e31ceab2bea11d7f4844ccc798
<code>FBundlePoint.cstT.eq_1</code>	thm	a0fcd53d02e27785302d863a3635c59859c33ca06a6f117db9cc80225fa7753
<code>FBundlePoint.derivT</code>	def	63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1
<code>FBundlePoint.derivT.eq_1</code>	thm	b3368a703919266cfc74c14b6f85e43d6c6f96467972bce1420abbe6b6d6e8059
<code>FBundlePoint.existsUnique_isFramedT</code>	thm	f602c4a06a1f754e26b273535cc054803334058b3c0d19b35d4499d478abb1a5
<code>FBundlePoint.isFramedT_iff_framingSystem</code>	thm	18fce3e0ae320ce7250aa0c3856a2a2bb00b455b589f195c9c3d3180cc662606

declaration	kind	SHA-256 of the statement
FBundlePoint.isUnit_det_of_tCoeffM_zero_eq_one	thm	a41826ffbacb1693561efac03bd5823ac9e65ec2a2e48879b46a2a357bd453d5
FBundlePoint.isUnit_det_tCoeffM_iff	thm	fe160d8ffba4f148895070785b114402e9aa2c8995249f3646a32f8a257e4f35
FBundlePoint.ofTCoeffs	def	932ce7e0a37797bb3f8e45963e2c50054ff7162ef375993683d1c7aae1eabbaf
FBundlePoint.ofTCoeffs.eq_1	thm	80b720e2b55dbf00ef4c111294053c370a0827f07cd3cd93224c72844b35b1f9
FBundlePoint.tCoeffM	def	937338866d69cdf7652ee0121613253529fb89ec8e4a538b4bfac1c326009300
FBundlePoint.tCoeffM.eq_1	thm	53b206d3e41294484a5d297ee88b1bd86d174fd17e37a8e715ff46a5a91aaacd
FBundlePoint.tCoeffM_add	thm	9f51cedc5697afd050b556f6df5cbf0c28fee277a0c1a1e0de388ddd4900ec18
FBundlePoint.tCoeffM_apply	thm	15fc69fc34c400f3a8748e18c2972c20358d8bb83cb3ee7f7b94866d0c640eff
FBundlePoint.tCoeffM_cstT	thm	5cb895daa1dfb733c9b080a2e608bba8b1866a24a1babf793cff3dc1df2c1fe6
FBundlePoint.tCoeffM_derivT	thm	8cfe07cee3293da91d3a9aeae09cde9c9345ea32f3ad422cf324c4b676ed8e19
FBundlePoint.tCoeffM_ext	thm	ff787f775938c35b1afc7566fc1dafd4baf30fc7e536ce23370eeebf9a00279c
FBundlePoint.tCoeffM_mul	thm	d7a52a514a93f6756db09a55bce5f9c270d9c3d6b1af152d7b7424c2092c3369
FBundlePoint.tCoeffM_ofTCoeffs	thm	b5ee7a5d4ceb52f45b233e6643cbe032c6461f5f5436ba1a8ebb97227629aaec
FBundlePoint.tCoeffM_uMul	thm	6d077fb8a8ece9216140969fb05a0d8041fdd8279e78ff87a4d4b9132e3b1a61
FBundlePoint.uMul	def	63b22e1e9b85ccc9c6721aa4785df23ef7ce10638842e2c0db8a0146368c52b1
FBundlePoint.uMul.eq_1	thm	7aac91d46e26581225d377f5f0b6f6509fd9d601e47eb3cd68d7349f455462b3

module	thm	def	role
<i>mathematics</i>			
BlockPartition	28	4	the same for arbitrarily many blocks, as in the paper
CompanionDiagonalization	31	6	diagonalization of that companion matrix by Hensel lifting
DirectionRing	41	6	both matrix lemmas over an arbitrary commutative $\mathbb{Q}$ -algebra
FBundlePoint	26	7	framed F-bundles over a point: the gauge system and its classification
FBundlePointPowerSeries	17	5	the same classification for matrices over $\mathbb{k}[[u]]$
FlatnessGauge	18	2	its instance for the framing operators; one step of the induction
FlatnessPropagation	11	4	the flatness equations that feed the matrix lemmas
FramingPropagation	3	2	propagation of a framing from one direction to another
GaugeCurvature	6	2	gauge covariance of the curvature, for abstract derivations
MatrixMultiDirection	27	4	both matrix lemmas over $\mathbb{k}[[t_i, i \in \sigma]]$
MatrixTDirection	10	2	the t -direction matrix lemma in one formal variable
MvIntegrability	6	1	the integrability obstruction with several formal directions
Nilpotency	8	0	the nilpotency condition and its consequence for framings
ProjectiveBundle	35	9	the companion matrix of a projective bundle; the corrected vanishing lemma
QFramingExistence	30	9	the framing recursion in a logarithmic direction
QFramingSeries	4	2	the same equation over $R[[q]][[u]]$
SpectralDecomposition	24	7	the spectral decomposition on the formal disk, order by order
SpectralPowerSeries	19	5	the same over $\mathbb{k}[[u]]$, with uniqueness
SplitTDirections	12	1	splitting of the t -directions along a block decomposition
Sylvester	18	3	Sylvester operators, coprime characteristic polynomials, two blocks
TDirectionExistence	9	3	existence for the t -direction system; well-posedness
TDirectionMv	8	1	the one-variable lemma as the $\sigma = \text{Unit}$ instance
TFramingExistence	29	9	the framing recursion in a t -direction
TFramingSeries	17	6	the same equation over $R[[t]][[u]]$
<i>cryptography</i>			
Crypto.Homomorphic	7	1	Paillier aggregation of encrypted shares
Crypto.Ledger	15	15	the statement ledger: completeness, hiding, soundness, binding
Crypto.Merkle	21	21	Merkle commitments: completeness and binding
Crypto.SecretSharing	16	4	the 24-of-24 key split
Crypto.Sigma	23	3	Pedersen commitments and the Schnorr Σ -protocol
Crypto.Stego	20	9	the LSB embedding and the reassembly of a deck

Table 1: The 30 modules of the library: 24 formalizing the mathematics of arXiv:2411.02266, 6 formalizing the publication scheme of §5. Counts are of kernel-extracted declarations, so they include the equation lemmas Lean generates for a definition.